The Metric Blind Spot in Deep Learning-Based Brain MRI Reconstruction: A Systematic Review

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Abstract

Objective: Deep learning accelerates brain MRI four- to tenfold, but learned models can erase a lesion or invent tissue, and pixel-averaged scores such as PSNR and SSIM miss both. We review whether evaluation practice could detect this blind spot. **Methods:** Following PRISMA 2020 we searched seven databases without date limit, included 263 studies (1995–2026), appraised them with QUADAS-2 and matched instruments, and synthesised narratively. Categories came from titles, abstracts and controlled vocabulary; every prevalence figure is a floor. Duplicate screening achieved substantial agreement (Fleiss' $\kappa = 0.877$); appraisal agreement was substantial (0.788; 0.390 where observable); extraction is unaudited. **Results:** Only 18 of 263 studies (6.8%) record a fidelity metric and a reader assessment on the same data, leaving the field's central proxy unmeasured. Reader studies mostly measure reader-versus-reader agreement, which is weak: fastMRI 2020 concordance was 0.457 and 0.386 (Kendall's W), improving only where SSIM had diverged. Erasing a 100 mm^3 lacunar infarct moves global PSNR about 0.03 dB under the stated error model. The corpus grew fivefold; reader assessment fell from 32% to 18%, recovering to 21%. Among the four largest families the most hallucination-associated (generative, 39%) is among the least reader-evaluated (11.3%); self-supervised is higher still (47%) with no readers. One study in twenty both releases code and runs readers; none evaluated a model observer; no named dataset covers acute stroke or

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haemorrhage. **Conclusions:** On these floors, evaluation practice cannot certify diagnostic safety. We derive five requirements safety-oriented evaluation must meet.

Keywords: Deep learning, Brain MRI reconstruction, Reconstruction safety, Hallucination, Image quality assessment, Systematic review

1. Introduction

Brain MRI is widely used because it can clearly distinguish different types of brain tissue and reveal subtle abnormalities that are difficult to detect with other imaging modalities. However, this image quality comes at the cost of long acquisition times. Unlike conventional imaging methods that capture an image more directly, MRI builds an image from a sequence of raw signal measurements (samples of spatial frequencies, known as k-space) collected over time. This prolonged acquisition reduces patient throughput and increases sensitivity to patient motion, which can introduce artifacts and degrade image quality (Zaitsev et al., 2015). To address these limitations, acceleration techniques such as parallel imaging, including Sensitivity Encoding (SENSE) (Pruessmann et al., 1999), and compressed sensing (Lustig et al., 2007) were developed to reduce the amount of data that must be collected. More recently, deep learning has enabled further acceleration by reconstructing high-quality images from incomplete MRI measurements, through cascaded convolutional networks (Schlemper et al., 2017), model-based unrolled networks (Hammernik et al., 2018; Aggarwal et al., 2018), generative adversarial networks (Yang et al., 2018), and score-based diffusion models (Song et al., 2021), at accelerations from fourfold to, in exploratory work, an order of magnitude (Radmanesh et al., 2022).

The reconstructed images look convincing, and that is the problem. Small changes in the input can produce large errors in the output (Antun et al., 2020; Gottschling et al., 2025). Four failure types recur across the studies we reviewed: strong priors smooth away faint structure; motion causes ghosting the model never saw in training (Sommer et al., 2020); a change of scanner or sampling degrades results in ways that in-distribution testing misses (Güngör et al., 2023); and generative models can invent plausible tissue (Zhou et al., 2024). The scores used to certify these methods, peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) (Wang et al., 2004), agree only partially with radiologist quality judgement even on conventional image degradations (Mason et al., 2020); and SNR, though reproducible across observers, tracks no-reference quality models only moderately (Yu et al., 2018), so an image can score well while the error that matters goes unrecorded. We call this the **metric blind spot**: standard image-quality metrics cannot catch these small but clinically serious errors, because a lesion of a few cubic millimetres barely moves an average taken over millions of voxels.

Parts of the field have responded, training quality metrics on radiologist rankings (Tang et al., 2025), attaching uncertainty maps to reconstructions (Khawaled and Freiman, 2024), or running reader studies that test an accelerated protocol against a reference one directly (Lang et al., 2024; Radmanesh et al., 2022). Causal and counterfactual methods can ask the most direct question, whether a lesion survives a controlled change of acquisition, but have mostly been applied to classification rather than reconstruction (Pawlowski et al., 2020; Castro et al., 2020). These responses share a root cause. Acceleration works because a learned prior fills in information the measurement does not contain, and that same filled-in information can hide a real finding or create a false one, while the field certifies the result with averaged pixel scores that cannot see either event.

To our knowledge no systematic review has assembled the evidence on this problem. Existing surveys cover architectures and benchmark performance (Liang et al., 2020; Montalt-Tordera et al., 2021); none asks how often conventional metrics miss clinically important errors, why they miss them, or what evaluation would have to look like to catch them. We address that through three questions:

RQ1 *How reliably do conventional fidelity metrics track radiologist diagnostic* *ratings in deep learning-based accelerated brain MRI, and how wide is the evaluation reporting gap in the literature?*

RQ2 *What underlying mathematical mechanisms cause standard metrics to overlook localized pathology erasure and generative hallucinations in deep learning-based reconstruction?*

RQ3 *What can each evaluation paradigm in current use establish about prior-induced diagnostic error in deep learning-based MRI reconstruction, and what requirements for safety-oriented evaluation follow from those limits?*

We contribute: (C1) a synthesis of what 263 primary studies report about agreement between pixel-wise metrics and radiologist judgement; (C2) four mathematical mechanisms that explain the divergence, with an exact identity bounding the PSNR cost of erasing a lesion; (C3) an analysis of *where* the evaluation gap comes from, showing that it co-occurs with a division of evaluation practice between research communities and with a benchmark infrastructure lacking acute pathology; and (C4) five requirements that safety-oriented evaluation must meet. The auditing framework these requirements motivate is developed and evaluated in a companion methods paper, not here.

Three boundaries should be stated at the outset, because they determine what the findings can and cannot be asked to support. First, this is a review of *evaluation practice*, not of reconstruction performance: we count what studies measured, not how well their methods performed, and nothing here estimates how often a deployed reconstruction actually erases a lesion. Second, the mechanisms in RQ2 are mathematical properties of the metrics, derived under a stated error model; they establish that a focal error *can* pass unrecorded, not that it commonly does. Third, prevalence counts rest on labels extracted from what studies report, so every one of them is a lower bound on practice rather than an estimate of it, and we treat the direction of that bias explicitly in Section 5.2. The contribution is a diagnosis of what the field's current instruments are capable of detecting, which is logically prior to, and considerably cheaper to establish than, a measurement of how often they fail.

2. Related Work

Four strands bear on this review, and the claim that none of them has assembled the evidence we assemble here is a claim about each in turn.

Surveys of reconstruction methods. Reviews of deep learning MRI reconstruction catalogue architectures, acceleration factors and benchmark accuracy (Liang et al., 2020; Montalt-Tordera et al., 2021; Knoll et al., 2020a; Kazerouni et al., 2023). They are organised by method, and their outcome variable is reconstruction fidelity against a fully sampled reference. Safety and failure modes appear as caveats in discussion sections rather than as an object of synthesis, and none of them tabulates what evaluation each primary study actually performed, which is the quantity this review extracts.

Task-based image quality assessment. The tradition that ought to have supplied the answer established, decades before deep learning, that image quality is meaningful only relative to a diagnostic task. Barrett and colleagues formalised the observer as part of the measurement: an image is good if an observer performing a specified task on it achieves a high figure of merit, typically the area under an ROC curve for a detection task, and the mathematical (ideal or channelised Hotelling) observer supplies a computable surrogate for the human when the task and image statistics are specified (Barrett et al., 1993; Barrett and Myers, 2004). ICRU Report 54 made the same argument the international reference framework for assessing medical imaging quality (International Commission on Radiation Units and Measurements, 1996). That framework already contains the concept the reconstruction literature lacks: a detectability index for a specified lesion, which is by construction sensitive to exactly the focal change a whole-volume L_2 score averages away. It has been carried into the dose- and protocol-optimisation literatures of other imaging modalities, but it has not crossed into deep learning MRI reconstruction: no study in this corpus evaluated a model observer, and our first requirement (R1, Section 4.7) is in substance a restatement of the task-based position for a field that has not adopted it. The reason it has not is worth naming: model observers require a specified signal and a statistical description of the background, and the null-space structure that makes learned reconstruction dangerous is precisely what makes those two ingredients hard to specify.

Failure-mode-specific work. Artifact detection, motion correction and robustness under distribution shift are each addressed in depth, but in isolation, by literatures that do not share an evaluation vocabulary (Esteban et al., 2017; Sommer et al., 2020; Güngör et al., 2023). Analyses of hallucination in tomographic reconstruction establish that prior-induced structure lives in the null space of the forward operator, so it is invisible to any measurement-space consistency check (Bhadra et al., 2021; Gottschling et al., 2025), and causal and counterfactual methods for medical imaging supply the machinery to intervene on image content and ask whether a finding survives (Pawlowski et al., 2020; Castro et al., 2020; Sanchez et al., 2022; Peng et al., 2026). Neither line has been connected to review-level evidence on how reconstruction is evaluated in practice.

What is new in the four mechanisms of RQ2, and what is not. Because Section 4 states four mechanisms formally, their provenance should be set out plainly rather than left for a reader to reconstruct. Two are elementary properties of the objective rather than findings: spatial error averaging follows from the definition of a mean over the image volume, and loss symmetry from the fact that squaring discards the sign of the residual. Two are established results we restate: reference dependence is standard in the image-quality literature, and the null-space account underlying the realism–reward trade-off is due to Bhadra et al. (2021) and Gottschling et al. (2025). What this review adds is not the mechanisms but their quantification and their assembly — the identity at Equation (2), which converts any lesion volume into the PSNR movement its erasure would cause, and the setting of all four against what a corpus of 263 studies is recorded as measuring. We claim the assembly and the bound, not the underlying results.

Benchmark and annotation infrastructure. The raw k-space resource the field benchmarks on is fastMRI (Knoll et al., 2020b), whose brain release followed the knee data described in that report, and an annotation layer is emerging on top of it: fastMRI+ adds expert pathology labels to the fastMRI brain and knee data (Zhao et al., 2022), and curated stroke-lesion datasets exist for segmentation (ISLES (Hernandez Petzsche et al., 2022), ATLAS (Liew et al., 2022)). None of these supplies raw k-space paired with acute vascular pathology, which is the combination reconstruction evaluation needs, and Section 4.5 shows that this is not an incidental gap but the binding constraint on the whole evaluation problem.

What no strand supplies is the join: a systematic account of how often the field measures metric–reader agreement, why the metrics fail when it does not, and what each evaluation paradigm can therefore establish. The four strands are not merely separate; they are held apart by what each can access. The survey and hallucination-analysis literatures work on datasets with raw k-space and no clinically verified pathology; the task-based tradition needs a specified signal and a labelled task, which those datasets do not carry; and the clinical reader literature has the pathology and the readers but no access to the raw data on which the mechanisms operate. Section 4.5 shows this is not an abstraction but a measurable division in the corpus itself. Supplementary Table S4 sets out each secondary study we identified against the questions asked here.

3. Methodology

This systematic review was conducted in accordance with the PRISMA 2020 statement (Page et al., 2021); the completed checklist, search vocabulary, extraction schema, and analysis scripts are available in the supplementary material and on the Open Science Framework (`https://doi.org/10.17605/OSF.IO/EUXA8`).

Protocol and registration status. The protocol, covering the research questions, eligibility criteria, search strategy, screening procedure, classification rubric and synthesis approach, was fixed in the project repository before screening began. In the terms of PRISMA 2020 item 24a, **this review was not prospectively registered and carries no registration number**:

PROSPERO's scope excludes reviews without direct clinical health outcomes, and no equivalent prospective registry covers methodological reviews of this kind. The Open Science Framework deposit cited above therefore timestamps the record but is retrospective, and we make no claim of prospective registration on it. Two consequences follow. An external record does not exclude undisclosed post-hoc analytic choices; and the two analyses we ourselves identify as post-hoc — the acceleration-factor extraction of Table 6 and the venue classification of Section 4.5 — are labelled as such wherever they appear.

Search strategy. Seven bibliographic databases (PubMed/MEDLINE, IEEE Xplore, Scopus, Web of Science, SpringerLink, OpenAlex, and Semantic Scholar) were searched between 23 and 24 July 2026, with no publication-date limit applied; the 1995–2026 span quoted elsewhere in this article is the publication range of the studies that were included, not a restriction imposed on the search. Queries were executed through each database's API rather than its web interface, so the complete Boolean match set was retrieved for every term combination rather than a ranked sample, explaining why the initial identification count is large. The strategy combined three required concept groups: neuroimaging (Block B1), reconstruction and acceleration (Block B2), and safety, failure modes, and evaluation outcomes (Block B3), with MeSH headings and wildcard truncation adapted per database (Supplementary Table S5). Table 1 details the executed query structure and hit breakdown across databases.

The breadth of this search is by design. Evidence of reconstruction failure is rarely a paper's stated subject: artifacts, hallucinations and metric–reader disagreements are most often reported in passing inside studies whose declared contribution is a new architecture, so a query tuned narrowly to safety vocabulary would miss much of this evidence. Clinical and engineering literatures also use distinct vocabularies — sequences and artifacts against inverse problems, unrolled optimization and generative priors — and neither database family indexes the other completely. We therefore tuned for sensitivity rather than precision and accepted the screening burden. The funnel from 27,327 records to

Table 1: Query structure executed in each database and record hit breakdown.

Database	Executed Query Structure	Controlled Vocabulary	Fields Searched	Hits (N)
PubMed / MEDLINE	(B1$_{ft}$[TIAB] OR B1$_{mesh}$) AND (B2$_{ft}$[TIAB] OR B2$_{mesh}$) AND (B3$_{ft}$[TIAB] OR B3$_{mesh}$) NOT (X[TIAB])	**Yes**: 9 MeSH headings OR-ed into the three blocks	Title/Abstract + MeSH	24,693
SpringerLink	(B1) AND (B2) AND (B3) NOT (X); quoted phrases, no truncation operators	No	Full record	1,598
IEEE Xplore	(B1) AND (B2) AND (B3) NOT (X); wildcard truncation retained	No	Metadata	778
Scopus	TITLE-ABS-KEY(B1) AND TITLE-ABS-KEY(B2) AND TITLE-ABS-KEY(B3) AND NOT TITLE-ABS-KEY(X)	No	TITLE-ABS-KEY	127
Semantic Scholar	(B1) + (B2) + (B3), where | denotes OR and + denotes AND	No	Title/Abstract	66
OpenAlex	title_and_-abstract.search:B1, title_and_-abstract.search:B2, title_and_abstract.search:B3	No	Title/Abstract	34*
Web of Science	TS=(B1) AND TS=(B2) AND TS=(B3) NOT TS=(X)	No	Topic (TS)	31
			Total identified records	**27,327**

Concept blocks combine as (B1) AND (B2) AND (B3) NOT (X). **B1** neuroimaging: "brain MRI", "cerebral MRI", neuroimag*, DWI, FLAIR, T1w, T2w, SWI; MeSH *Brain*, *Magnetic Resonance Imaging*, *Neuroimaging*. **B2** AI and accelerated reconstruction: "MRI reconstruction", "accelerated MRI", "undersampled reconstruction", "k-space reconstruction", "unrolled network", GAN, "diffusion prior"; MeSH *Deep Learning*, *Machine Learning*. **B3** safety, artifacts and evaluation: hallucinat*, "lesion suppression", reconstruction artifact*, PSNR, SSIM, "reader study"; MeSH *Artifacts*, *Diagnostic Errors*, *Reproducibility of Results*. **X** exclusion: "animal model", "case report", "in vitro". * denotes truncation. Complete term lists, MeSH headings and per-database syntax are given in Supplementary Table S5. Searches were executed between 2026-07-23 and 2026-07-24, with no publication-date limit. *OpenAlex returned 50 raw hits; 34 passed record-completeness filtering.

263 studies is the intended signature of that choice: the burden falls on initial screening, which reviewers can absorb, rather than on coverage, which no downstream step could recover.

Eligibility and selection. Records were managed in the review's own Python pipeline rather than a commercial screening platform; the scripts, the per-record decision files and the per-database query logs are released, so each step below is re-executable. Deduplication by exact identifier matching (PMID/DOI) and string/TF-IDF similarity (Jaro-Winkler $\geq$ 0.97, cosine $\geq$ 0.95) removed 403 duplicates (1.47%), leaving 26,924 unique records. That rate is low enough to suggest a matching failure but is structural: PubMed's MeSH arm returned 24,693 records and the six engineering arms 2,634 between them, and sets that barely overlap leave little to deduplicate. This is not a claim of completeness, and one failure is on record: peer review identified an arXiv preprint and its journal version that both survived to the included set, carrying different DOIs, no shared PMID, and titles differing by terminal punctuation — below the thresholds above. It was removed at revision, taking the corpus from 264 reports to 263 studies (Section 4). Automated matching will not catch a preprint–journal pair whose metadata diverges in this way, and the honest inference is that a residual duplicate rate of the same order cannot be excluded.

Three reviewers (D.T.M., T.V.P., T.N.T.D.) screened records against the pre-specified PICOS criteria (Table 2) at both stages, under a primary-screen-plus- adjudication design: each record received one primary label of *include*, *exclude*, *flagged* or *background only*, and every record labelled *flagged* went to a second reviewer for adjudication. Labelling was deliberately conservative, so any record with an unclear PICOS dimension, or whose text we could not obtain, was flagged rather than excluded. Title and abstract screening advanced 1,149 records to full-text assessment, of which 109 were labelled *include* on the abstract alone and 1,040 *flagged*; both labels carried the record forward, and the distinction records only whether the abstract was sufficient to decide. A further 25,573 were excluded and 202 set aside as background. Full texts were pursued through institutional and title-level library subscriptions, open-access repositories, preprint servers, PubMed Central and inter-library request until no route remained: 1,068 of the 1,149 (93.0%) were obtained and assessed, and the 81 that could not be obtained are broken down by recorded reason with the selection results (Section 4).

The three reviewers divided the 1,068 obtained reports between them under the same primary-screen-plus-adjudication design used at title and abstract: each report received one primary full-text label, and a report was escalated to a second reviewer wherever the primary label was *flagged*, wherever an inclusion decision rested on a borderline PICOS dimension, or wherever the reviewer requested a second reading. Escalated reports were resolved by consensus discussion between the two assessors, with the third reviewer consulted where the two did not converge. The escalation volumes this produced are reported below, and the per-report labels, escalation marks and reasons are released in the decision ledger. We state the design in this form because it determines what can be computed from the ledger: one primary label per report means no chance-corrected inter-rater coefficient is recoverable from full-text screening, which is why the separate criterion-reproducibility exercise below was run. In total, 263 primary studies met all inclusion criteria upon full-text verification (two additional reports were excluded as duplicate publications). Full-text exclusions ($n = 785$) were categorized by pre-specified PICOS criteria: wrong intervention ($n = 326$), out of scope ($n = 256$), wrong outcome ($n = 98$), wrong population ($n = 93$), wrong study design ($n = 4$), non-English ($n = 1$), and non-primary articles ($n = 7$). A further 18 reports were retained as background material rather than as included studies, so the 1,068 assessed reports resolve as 263 included, 785 excluded on PICOS grounds, 2 excluded as duplicate publications and 18 retained as background. The complete decision ledger is released as `fulltext_screening_decisions.csv`.

Publication year. Every analysis partitioning the corpus by time uses the year of the version of record as indexed by the publisher — the issue year for journal articles, the proceedings year for conference papers — rather than first online posting or preprint deposit. The rule matters at era boundaries, and we adopt the indexed year because it is what a reader can verify from the reference list, Crossref and PubMed without online-first metadata, which is unevenly available across the 87 venues represented. The same value is recorded in the `year` field of every bibliography entry and in the `publication_year` column of `included_characteristics.csv`, so the era counts in Table 9 are recomputable from either. One consequence should not be read as an inconsistency: the searches ran on 23–24 July 2026, yet 41 of the 263 included studies (15.6%) carry a 2026 issue year, being articles published online ahead of print before the searches ran but assigned by the publisher to a 2026 issue.

Screening agreement, and what it does and does not cover. The primary screen described above records one label per record, so no chance-corrected coefficient is recomputable from those decisions alone. We therefore ran a separate duplicate-screening exercise. A random sample of 214 records already assessed at title and abstract was re-screened independently by all three reviewers, blind both to the recorded decision and to one another. Reviewers judged the four PICOS dimensions from title, journal and abstract, opening the DOI link only where the abstract was insufficient, and returned one of three labels per record: include, exclude or unsure. The sample was drawn without stratification; the drawn records, the three label sets and the agreement computation are released, so the sample is auditable record by record. All 642 cells were completed and no record was dropped for missingness.

Across all 214 records Fleiss' $\kappa = 0.877$ (95% bootstrap CI 0.823–0.925; 10,000 resamples, seed 20260821), with observed agreement $\bar{P} = 0.933$ against $\bar{P}_e = 0.455$ expected by chance. The three reviewers agreed unanimously on 194 of 214 records (90.7%), and pairwise Cohen's κ ran from 0.845 to 0.931. One sensitivity analysis is worth reporting because it is the one a reader is most likely to run. Restricting to the 193 records on which no reviewer recorded *unsure* raises Fleiss' κ to 0.943 (95% CI 0.900–0.979). That complete-case figure discards 21 records, and it discards precisely the records at least one reviewer could not decide, so it overstates agreement on the population the coefficient is meant to describe. We report 0.877 as the estimate and 0.943 only as the value obtained when the hard cases are removed. Two qualifications travel with that figure and should not be dropped. First, it measures agreement on the screening decision, not on the extraction labels from which every prevalence count in this article is derived; those remain unaudited for reliability, and the only error rate we can quote for them is the measured 4/61 (6.6%) on the design field (Section 5.2). Second, the conservative labelling policy routed almost the whole full-text workload to adjudication in any case — 1,040 of the 1,149 records advanced at title and abstract were flagged rather than included outright — so a high coefficient is evidence that the criteria were applied consistently, not that the borderline records were easy to classify.

What the primary decision records themselves support, and what we therefore also report, are the escalation volumes that policy produced. At full text, 84 reports were escalated to a second reviewer for adjudication and 29 of those were included; a further 77 reports carry a mandatory human-re-inspection mark in the released files, of which 40 were included after that review. Screening recall is bounded by the search strategy and by the criteria as applied rather than estimated against an independent reference set of known-eligible records, since no such set exists for this question; Section 5.2 treats both points as limitations. The duplicate-screened sample, both label sets and the agreement computation are released as `kappa_human_results.csv`, alongside the full decision ledger in `fulltext_screening_decisions.csv`.

Data extraction. Reviewers extracted study characteristics using the pre-specified form (Supplementary Table S6), capturing method family, reported failure modes, evaluation paradigm, field strength, pulse sequence, reference standard and reproducibility indicators, with a verbatim supporting quotation for every assigned value. Method, failure-mode and evaluation categories were assigned from each record's title, abstract and controlled vocabulary against the vocabulary fixed in advance (Supplementary Table S5); that deliberately conservative scope is why every prevalence count here is a lower bound rather than an exhaustive tally. Field strength and pulse sequence, typically stated only in acquisition subsections, were additionally taken from full text where available. We did not contact investigators: a variable not stated in the source was recorded as *not reported* and never inferred, which is why the 54 studies without a stated field strength divide into 50 whose full text does not state it and 4 assessed on abstract or partial text.

Table 2: PICOS elements and their application in this review.

Element	Definition	Application in this Review
Population	Target imaging modalities, clinical domains, and scanner physics	Human brain MRI acquisitions (acute ischemic stroke, neurovascular lesions, intracranial haemorrhages, tumours, and structural T1w, T2w, FLAIR, DWI scans across 1.5T, 3T, and 7T field strengths).
Intervention	AI-based and accelerated reconstruction methods, and their safety/fidelity evaluation	Deep learning and accelerated reconstruction methods (unrolled physics-based networks, GANs, diffusion priors, self-supervised, denoising, super-resolution), together with methods for evaluating their fidelity, artifacts, robustness, or failure modes.
Comparison	Baseline metrics, observers, and reference standards	Conventional pixel-wise metrics (PSNR, SSIM, NRMSE), perceptual/learned metrics (LPIPS, FID), model observers, fully sampled reference images, or radiologist reader assessments.
Outcome	Reconstruction quality, fidelity, and failure characterization	Image-quality/fidelity measures, artifact and failure-mode characterization, metric–radiologist agreement, uncertainty estimates, robustness to distribution shift, and diagnostic/observer outcomes.
Study Design	Methodological criteria and empirical validation settings	Primary empirical studies, algorithmic benchmarking, counterfactual interventional stress-testing, and human or model observer evaluations.

Quality appraisal, and the reader/algorithmic partition. Methodological quality was evaluated with three design-matched instruments: QUADAS-2 (Whiting et al., 2011) across its four domains for diagnostic accuracy and radiologist reader studies; a reporting-transparency check against STARD principles (Bossuyt et al., 2015), reported descriptively rather than scored, since STARD yields no risk-of-bias rating; and, for algorithmic benchmarking studies, a reproducibility and robustness appraisal covering code and data availability, use of a named public benchmark, whether the reference standard was a fully sampled acquisition, and whether robustness to acquisition or distribution shift was reported. ROBINS-I (Sterne et al., 2016) was not applied, since few included studies are the non-randomised interventional designs it was built for.

The partition between the two instruments is not the one the extraction labels originally implied, and the correction matters for every reader-share statistic here. Extraction recorded 61 studies as observer or reader designs; three-reviewer appraisal of their full text found four containing no human reader at all, each using the language of observer evaluation for an automated surrogate. Those four were reclassified as algorithmic, so **the verified reader subset is** 57 **and the algorithmic set is** 206, and $57 + 206 = 263$ closes the corpus exactly. All 57 were appraised by all three reviewers across two rounds, 39 in the first and 18 in the second, so every QUADAS-2 tally rests on 57 studies and $57 \times 7 = 399$ domain judgements. Every reader-derived count in this article uses 57, and the released per-study file encodes the same partition with the reclassification recorded per study. One further study (S001) carries an appraisal note that a human reader could not be *confirmed*, only abstract-level text being available; it is retained in the reader subset, since non-ascertainment is not a finding of absence, and it is flagged in the deposit.

Appraisal agreement. The three reviewers rated every applicable domain Low, Unclear or High independently, each judgement carrying a verbatim supporting quotation, before joint review to harmonize thresholds. Five first-round studies were later re-appraised on recovered full text (below), so their 35 judgements no longer rest on three independent ratings and are excluded from the coefficient. Over the remaining 52 studies and 364 judgements Fleiss' $\kappa = 0.788$ (95% bootstrap CI 0.739–0.833; 10,000 resamples, seed 20260821). That figure averages two rounds that behaved differently, and the difference is more informative than the average. The 34 first-round studies retaining three independent ratings were unanimous on all 238 of their judgements, so they contribute $\kappa = 1.00$ by construction and carry no information about latent reliability. The 18 of the second round were unanimous on 60 of 126 (47.6%), and across those 126 Fleiss' $\kappa = 0.390$ (95% CI 0.293–0.484) — fair on the Landis–Koch bands, with an interval reaching into moderate. The procedure was identical across rounds; the subsets were not, the second round comprising studies the first pass deferred as not straightforward. Two readings remain open — that those studies report their methods less completely and are genuinely harder to rate, and that the first round was less independent in practice than in design — and these data do not separate them.

The disagreement has a structure worth recording rather than smoothing. Across the 126 second-round judgements one reviewer rated Low 33.3% of the time and another 61.9%, a difference of threshold rather than of reading, which is why κ sits at 0.390 while raw agreement is 63.5%. Six judgements split three ways with no majority; all six were recorded as *Unclear*, on the ground that a panel unable to converge is itself evidence the report did not support a confident judgement, which is what *Unclear* encodes in QUADAS-2. The rule was applied to those six and no other case, and the per-reviewer labels are released so a reader can apply a different rule and recompute. Section 5.2 states what this does and does not establish. Table 3 summarizes corpus-level quality indicators and the QUADAS-2 domain judgments.

Table 3: Corpus-level provenance and methodological quality indicators across the 263 included studies, with QUADAS-2 domain-wise risk of bias and applicability concerns for the full appraised diagnostic/reader subset ($n = 57$, rated across seven domains; independently by all three reviewers except for the five studies re-appraised on recovered full text). The 39 studies of the first appraisal round and the 18 of the second are pooled here; the remaining 206 studies were appraised on reproducibility and robustness instead ($57 + 206 = 263$). Six of the 399 domain judgements split three ways with no majority and are recorded as *Unclear* (Section 3).

Indicator / Domain	**Value / Risk of Bias Breakdown**
Publication period	1995–2026; 229/263 (87.1%) published in 2020 or later, and 174/263 (66.2%) in 2023 or later, indicating a predominantly recent evidence base.
Basis of assessment, current	Full text available for 258/263 (98.1%); partial or truncated full text for 1/263 (0.4%); title, abstract and MeSH only for 4/263 (1.5%).
Basis of assessment, at appraisal	The first appraisal round was made on the text obtainable at the time, which for 18/263 studies was abstract and MeSH only (6 of them within the QUADAS-2 subset). A later library sweep recovered full text for 14 of those 18 (including 5 of the 6), and all 14 have since been re-appraised on the recovered full text, so appraisal basis now stands at 257/263 full text, 2/263 partial text and 4/263 abstract and MeSH only — within the QUADAS-2 subset, 56/57 on full text and 1/57 (S001) on abstract-level text. The two rows now differ by a single study (S200), appraised on truncated text whose full version has since become available.
Leading venues	The eight most frequent of the 87 distinct venue titles represented in the corpus, accounting for 128/263 (48.7%) of included studies: *Magnetic Resonance in Medicine* (40), *IEEE Transactions on Medical Imaging* (17), *AJNR* (15), *Magnetic Resonance Imaging* (14), *JMRI* (12), *Scientific Reports* (12), *NeuroImage* (9), *Medical Image Analysis* (9).

(Table 3 continued)

Predominant design	Algorithmic / methodological reconstruction studies ($n = 206$), alongside a verified observer/reader subset ($n = 57$) (Radmanesh et al., 2022; Lang et al., 2024). Extraction recorded 61 reader designs; appraisal of their full text found 4 with no human reader, and those 4 were reclassified as algorithmic ($202 + 4 = 206$; $61 - 4 = 57$). Of the 57: 34 are appraised and verified by all three reviewers; 18 were appraised by all three in a second round and await final adjudication of 6 split domain judgements; and 5 (S004, S020, S027, S034, S115) were re-appraised by the reviewers on full text recovered after the first round, with domain judgements verified against the source text. Every reader-derived statistic in this article uses 57.
Reference standard	Expert reader consensus (70), an unspecified stated ground truth (62), a fully sampled acquisition (69), a conventional or standard-of-care acquisition (15); not stated in 47/263 (17.9%).
Reproducibility	Code availability stated in 97/263 (36.9%); a named public dataset or benchmark used in 109/263 (41.4%), most commonly fastMRI, HCP, ADNI and IXI. Availability is variable and remains an area for improvement.
Sampling and setting	Sampling frame not stated in 95/263 (36.1%); recruitment centre not stated in 174/263 (66.2%). Ethics approval or informed consent reported in 159/263 (60.5%).
QUADAS-2 Risk of Bias ($n = 57$ Reader / Diagnostic Accuracy Studies)	
Patient Selection	Low: 15/57 (26.3%); Unclear: 19/57 (33.3%); High: 23/57 (40.4%). The weakest domain: sampling is predominantly purposive or pathology-enriched, and most of the appraised studies state no enrolment rule.
Index Test (AI Reconstruction)	Low: 26/57 (45.6%); Unclear: 9/57 (15.8%); High: 22/57 (38.6%). Low risk falls below half once the second round is included. High ratings arise where readers scored accelerated reconstructions with the reference visible, or where test parameters were set using reference-derived labels.
Reference Standard	Low: 26/57 (45.6%); Unclear: 15/57 (26.3%); High: 16/57 (28.1%). High ratings arise where no artifact-free reference exists, or where lesion truth is defined by the comparator images themselves (incorporation bias). Reading the recovered full text moved four of the re-appraised studies from Low to High here, each because the comparator images, or software applied to them, supplied the truth against which they were scored.
Flow and Timing	Low: 45/57 (78.9%); Unclear: 7/57 (12.3%); High: 5/57 (8.8%). The strongest domain, since retrospective undersampling of the same raw k-space gives a zero interval between index test and reference.
QUADAS-2 Applicability Concerns ($n = 57$)	
Patient Selection	Low: 36/57 (63.2%); Unclear: 4/57 (7.0%); High: 17/57 (29.8%), chiefly healthy-volunteer, lesion-free, or non-clinical cohorts.

(Table 3 continued)

Index Test	Low: 30/57 (52.6%); Unclear: 4/57 (7.0%); High: 23/57 (40.4%), where the evaluated method is image-domain post-processing, cross-field synthesis, harmonization, or a quality index rather than reconstruction from under-sampled k-space.
Reference Standard	Low: 39/57 (68.4%); Unclear: 3/57 (5.3%); High: 15/57 (26.3%).

Across all 228 risk-of-bias judgements: 112 (49.1%) Low, 50 (21.9%) Unclear, 66 (28.9%) High. Across all 171 applicability judgements: 105 (61.4%) Low, 11 (6.4%) Unclear, 55 (32.2%) High. Over the 364 judgements that rest on three independent ratings Fleiss' $\kappa = 0.788$ (95% CI 0.739–0.833), but agreement is not uniform across the two appraisal rounds: the 238 retained first-round judgements were unanimous, the 126 of the second round reach $\kappa = 0.390$ (95% CI 0.293–0.484); the 35 judgements of the 5 re-appraised studies are excluded, having been conducted subsequently on recovered full text. Section 3 gives the decomposition and Section 5.2 what it bounds. The high proportion of Unclear ratings reflects unreported blinding and sampling procedures, compounded by the one first-round study for which only abstract-level text is available. Six second-round judgements split three ways with no majority and are recorded as Unclear under the rule stated in Section 3. Per-study ratings, each with the verbatim supporting quotation, are released as `included_characteristics_supplement.csv`.

Derived venue classification. The analysis in Section 4.5 groups studies by publication community. That grouping is post-hoc, was not part of the fixed protocol, and is stated in full in Supplementary Table S13 so it can be reproduced or contested. Each study's journal title, and nothing else about the study, is matched case-insensitively against three ordered keyword lists — engineering and methods, then general neuroscience and neuroimaging, then clinical imaging — with anything unmatched left *unclassified* rather than forced into a group. The ordering is load-bearing, since titles such as *Medical Image Analysis* would otherwise match two lists, and it is applied identically to all 263 studies. The rule as executed and the resulting per-study assignment are released as `5_Code/venue_classification.py` and `venue_community_classification.csv`.

Synthesis. Due to heterogeneity in acquisition protocols, reconstruction architectures, datasets, and outcome measures, evidence was synthesised narratively in accordance with SWiM guidelines (Campbell et al., 2020). Studies were structured around the primary research questions (RQ1–RQ3) and grouped by reconstruction method family, field strength, pulse sequence, and failure mode. Quantitative values reflect named primary studies rather than pooled meta-analytic estimates; no effect measure is pooled, and no meta-analytic model is fitted. Percentages are reported to one decimal place and rounded half-up throughout, so 9/80 prints as 11.3% and not 11.2%. Sensitivity analysis was performed by recalculating headline prevalence counts on the 258 studies verified against complete full text (Section 5.2).

4. Results

All counts and prevalence statements in this section describe the 1,068 reports assessed in full and the 263 included studies; because 81 candidate reports remain unretrievable, they are lower bounds on the literature rather than estimates of it.

4.1. Selection and corpus

From 27,327 records identified across seven databases, multi-tier deduplication removed 403 duplicates, leaving 26,924 unique records screened at title and abstract. Of these, 1,149 proceeded to full-text eligibility, and full texts were obtained for 1,068 of them (93.0%; Section 3). Each of the 81 reports still out of reach was checked against its publisher platform and assigned a recorded reason: 53 sit behind title-level subscriptions our institution does not hold, across nineteen publishers; 13 are Springer conference chapters sold only per chapter; 4 are ISMRM proceedings abstracts issued without a PDF; 4 resolve to a whole proceedings volume or to front matter rather than a single report; and 7 carry no DOI and no locatable source. Supplementary Table S14 gives the per-publisher breakdown. Access difficulty is itself informative, though it bounds the missing evidence rather than characterising it. Of the 341 assessed reports obtainable only through title-level subscriptions and equivalent barriers, 294 (86.2%) proved ineligible on the same PICOS criteria, 2 were retained as background and 45 were eligible ($294 + 2 + 45 = 341$; the 45 resolve to 44 studies, one being a preprint adjudicated a duplicate). Raw non-retrieval counts therefore overstate the missing evidence by roughly sevenfold. But those 45 were concentrated in subscription clinical journals, and Section 4.5 shows they differ systematically from the open literature in exactly the variable this review measures, so 86.2% is an upper bound on what non-retrieval costs us, not a demonstration that it costs us nothing. Of the 1,068 assessed reports, 787 were excluded (785 on PICOS grounds, 2 as duplicate reports of included studies) and 18 recorded as background, leaving **263 included primary studies** spanning 1995–2026 (Fig. 1). Each included study was verified against the fullest text obtainable for it: complete full text for 258, partial text for 1, and title, abstract and controlled vocabulary for 4 (Section 3). Their aggregate characteristics are given in Table 4, and their thematic distribution across the research questions is summarized in Supplementary Table S7; all 263 are listed, with full references, in Supplementary Tables S1–S3.

Corpus growth is recent: 34 studies appeared in 1995–2019, 55 in 2020–2022 and 174 in 2023–2026.

4.2. RQ1: the metric–reader comparison is almost never made, and where readers appear they agree only partially

Across the 84 studies reporting objective fidelity metrics and the 57 verified reader studies, the comparisons that actually exist show agreement between fidelity metrics and radiologist assessment that is partial and condition-dependent (Table 5). The size of that evidence base should be stated in the text and not left to the table caption: only two of the four rows in Table 5 were collected on reconstructions of undersampled k-space. Tang et al. (2025) collected radiologist rankings on synthetically corrupted image pairs, and Sommer et al. (2020) on clinical scans corrupted by real patient motion rather than by undersampling. The table that carries RQ1 therefore rests on two on-target rows, which is the honest size of the direct evidence for the metric–reader comparison in this setting. In the fastMRI 2020 challenge's reader phase, six radiologists ranked submissions for quality of pathology depiction; their concordance, by Kendall's coefficient, was 0.457 in the 4× track and 0.386 in the 8× track, rising to 0.781 only in the transfer track, and the organisers report that concordance improved as SSIM scores diverged, which is to say the metric separated methods

Table 4: Characteristics of the 263 included studies.

Demographic / Method Dimension	**Category Breakdown**	**(n)**	**Percentage / Key Methodological Notes**
Publication Period	1995–2019	$n = 34$	12.9% (Classical CS/PI & early CNNs)
	2020–2022	$n = 55$	20.9% (Unrolled physics & GAN priors)
	2023–2026	$n = 174$	66.2% (Diffusion priors & 3D/4D foundation models)
Reconstruction Paradigm *(multi-label;* 339 *labels recorded)*	Compressed Sensing / Parallel Imaging	$n = 93$	Iterative sparsity & coil sensitivity encoding
	Generative (GAN/Diffusion/VAE)	$n = 80$	Adversarial perception & posterior sampling
	Motion Correction	$n = 57$	Intra-scan motion artifact suppression
	Physics-based Unrolled	$n = 37$	Explicit k-space data consistency layers
	Denoising	$n = 30$	Image-domain noise suppression
	Super-resolution	$n = 25$	Resolution enhancement post-acquisition
	Self-supervised / Unsupervised	$n = 17$	Training without fully sampled reference scans
MRI Pulse Sequences *(multi-label;* 490 *labels)*	T1-weighted (T1w)	$n = 163$	High-resolution anatomical cortical profiling
	T2-weighted (T2w)	$n = 122$	Ventricular & CSF pathology assessment
	FLAIR	$n = 85$	Hyperintense peri-ventricular lesion detection
	Diffusion-Weighted (DWI)	$n = 72$	Acute ischemic lesion (DWI core) evaluation
	SWI / T_2^*-weighted	$n = 48$	Punctate cerebral microbleed hypointensity voids
Magnetic Field Strength *(single-label; predominant field where several are used)*	3.0 Tesla (3T)	$n = 153$	58.2% (Primary clinical scanner benchmark)
	7.0 Tesla (7T)	$n = 35$	13.3% (Ultra-high-field research acquisitions)
	1.5 Tesla (1.5T)	$n = 21$	8.0% (Routine clinical throughput imaging)
	Not stated	$n = 50$	19.0% (Full text obtained; no field strength stated)
	Not ascertainable	$n = 4$	1.5% (Only abstract or partial text available)
Evaluation Paradigm *(multi-label;* 177 *labels recorded across* 144 *studies)*	Pixel-wise (PSNR/ SSIM/NRMSE)	$n = 84$	47.5% of recorded (legacy fidelity reporting)
	Observer / Reader Studies	$n = 57$	32.2% of recorded (task-based reader assessment; verified subset)
	Uncertainty Estimation	$n = 18$	10.2% of recorded (posterior variance maps)
	Learned / Perceptual	$n = 18$	10.2% of recorded (rank-trained metrics)

Co-reporting. Of the 144 studies with a recorded evaluation paradigm, 28 report more than one. Critically, only **18** studies in the entire corpus (6.8% of 263; 12.5% of the 144 recorded) report *both* a pixel-wise fidelity metric *and* a reader assessment on the same data, which is the minimum precondition for computing metric-versus-reader agreement on which safety claims implicitly depend. See Section 4.

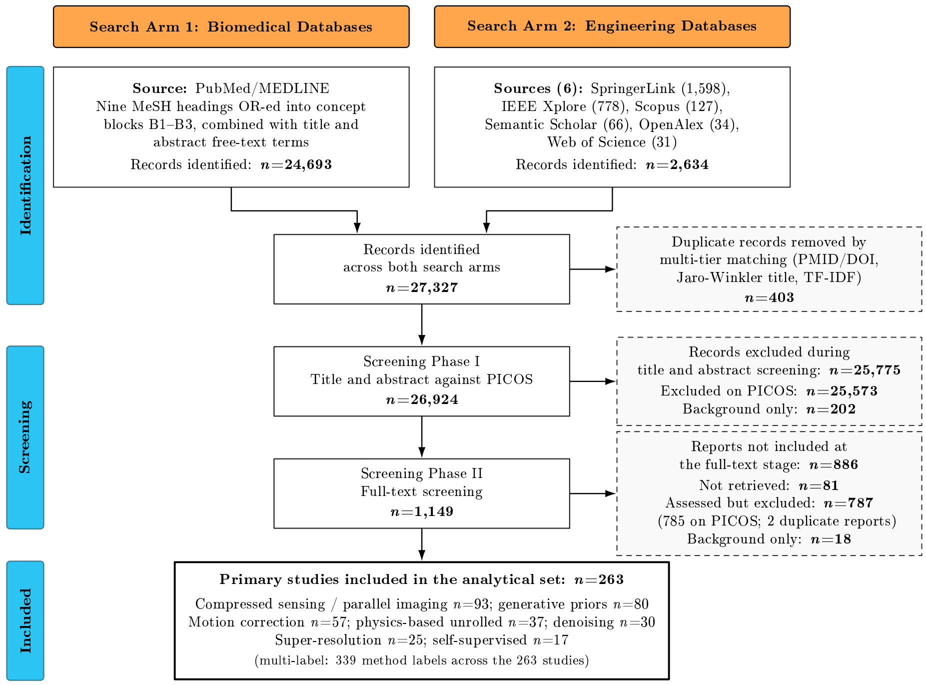

Figure 1: PRISMA 2020 flow diagram for study selection across the two search arms. Multi-tier deduplication reduced 27,327 identified records to 26,924 unique records; title and abstract screening advanced 1,149 candidates. Full texts were obtained for 1,068 of them (93.0%); 81 could not be retrieved. Following PRISMA 2020, exclusion and background retention are recorded as distinct dispositions: of the 1,068 reports assessed in full, 787 were excluded (785 on explicit PICOS grounds and 2 as duplicate reports of included studies) and a further 18 were retained as background material rather than excluded, leaving 263 primary studies in the analytical set ($263 + 787 + 18 = 1{,}068$).

reliably only where their outputs already differed grossly (Muckley et al., 2021). When radiologist preference is instead used as the training signal, purpose-built quality networks predict expert rankings at 75.2% ± 1.3% and 79.2% ± 1.7%, against radiologists' own interobserver agreement of 61.5% ± 5.5%; the same study finds rank-based reading far more reliable than Likert scoring (70.4% versus 25% agreement between radiologists) (Tang et al., 2025). Reader tolerance of acceleration is real but bounded: in a retrospective evaluation spanning accelerations up to 100×, 94% of volumes (65 of 69) were rated sufficient for potential screening use at up to 14× (interreader intraclass correlation 0.875), with the threshold for full diagnostic quality lower still (Radmanesh et al., 2022). The one direct clinical observation of the failure this review is about comes from an emergency-department and inpatient comparison, and it is a single case. Across 66 paired examinations in which a two-minute ultrafast protocol (T1w, T2/T2*, FLAIR and DWI, machine-learning-assisted reconstruction) was read against a ten-minute reference protocol, two blinded neuroradiologists agreed on the main clinical diagnosis in 65 of 66 cases (98.5%), rated overall diagnostic quality no different between protocols ($P > .05$), preferred the reference protocol for image noise and geometric distortion ($P < .05$) and preferred the accelerated protocol for reduced motion artifact ($P < .01$) (Lang et al., 2024). The single discrepant case is the one that matters here: a punctate focus of restricted diffusion, read as possible acute-to-subacute infarct on the reference DWI, was less conspicuous on the accelerated DWI, which the authors attribute to a combination of image-quality difference and section positioning, the lesion being smaller than the section thickness (4 mm reference against 5 mm accelerated) and so subject to partial-volume contamination. They draw the inference themselves, that the protocol's sensitivity to small findings is an open question. One case in 66 is not a prevalence estimate and we do not offer it as one. It is a worked instance of the pattern this review predicts: an aggregate quality comparison that finds no difference, and inside it a focal finding that a reader saw and the aggregate did not. Where readers assess artifacts directly, blinded reading detects what reference-based metrics cannot express: a two-neuroradiologist blinded study of 28 clinical scans with real patient motion found significant artifact-score reductions after network correction ($p < .03$), a setting in which the motion-corrupted reference invalidates PSNR and SSIM by construction (Sommer et al., 2020). Motion also interacts with the sampling scheme itself, appearing as ghosting under Cartesian trajectories and as streaking or blurring under radial ones, so the visual form of the same physical corruption differs by protocol (Schauman et al., 2026). The reference pipeline is not neutral either: conventional SWI homodyne phase filtering has been shown to create punctate artifactual microhaemorrhage-like foci, so processing can manufacture findings as well as remove them (Li et al., 2015).

No included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment for deep learning-based brain reconstruction, and none evaluated a model observer. The second statement needs a boundary that the first does not. Our search vocabulary (Table 1, block B3) contains reader-study and metric terms but no model-observer vocabulary, so a study that used a channelised Hotelling or non-prewhitening observer and described it only in those terms could have been missed at retrieval. What we can say is that none of the 263 studies retrieved, screened and extracted here reports one, and that the term does not appear in any extracted evaluation label; whether the true field-wide count is zero or merely very small, it is not a paradigm this corpus can be said to use. That absence is itself the RQ1 finding: only 18 of 263 studies (6.8%, a screening-level lower bound; Section 3) record a fidelity metric and a reader assessment on the same data, so the field's central proxy relationship, that high fidelity scores imply diagnostic adequacy, is largely unmeasured rather than merely disputed.

Table 6 reports how evaluation practice varies with the acceleration factor actually stated in each study. It is built only from what the corpus records: the acceleration factor was extracted post hoc from the full texts of all 263 included studies with a verbatim supporting quotation for every value, and 63 studies (24.0%) state one. Studies are grouped by the highest factor at which they evaluate their method. The table deliberately carries no fidelity threshold, reader-score difference or metric–reader rank statistic per acceleration band, because no included study reports one: that absence is the finding of Section 4, not an omission from the table.

4.3. RQ2: four mechanisms explain the divergence

Four mechanisms account for why the metrics miss what the readers see. Table 7 states each one formally alongside the clinical failure it permits, and Supplementary Table S8 gives the fuller derivations and evidence sources.

Spatial averaging. Conventional L_2 metrics average residual error uniformly over all $H \times W \times D$ voxels, so the total decomposes over lesion and background regions:

Table 5: What the reader studies in this corpus actually measured about metric–reader agreement (RQ1). Every value is quoted from the named study, with the reported coefficient and its definition stated. No included study reports a metric–reader rank correlation for deep learning-based brain reconstruction; the rows below are the closest measurements that exist, and the second column states for each row what the readers were actually shown, because two of the four rows were not collected on reconstructions of undersampled k-space.

Study	**Readers and design**	**Comparison performed**	**Reported quantity (as defined in the study)**
fastMRI 2020 challenge (Muckley et al., 2021)	6 radiologists ranking submissions for quality of pathology depiction	Radiologist concordance across accelerations; SSIM-based ranking vs reader ranking	Kendall's $W = 0.457$ ($4\times$), 0.386 ($8\times$), 0.781 (transfer track); concordance improved as SSIM scores diverged
Tang et al. (2025)	7 radiologists, rank-based reading; 2 also Likert scoring. **Rankings were collected on synthetically corrupted image pairs, not on reconstructions of undersampled k-space**	Quality networks trained on radiologist rankings vs MSE/SSIM as optimisation targets	Ranking accuracy $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ with reference image; radiologist interobserver agreement $61.5\% \pm 5.5\%$; ranking vs Likert agreement 70.4% vs 25%
Radmanesh et al. (2022)	Retrospective; neuroradiologist scoring of diagnostic quality at accelerations up to $100\times$	Reader-assigned acceleration thresholds for diagnostic and screening quality	65/69 volumes (94%) sufficient for potential screening at $\leq 14\times$; interreader ICC 0.875; mean absolute interreader difference in threshold acceleration factor $0.7\times$ (level 1) and $5.4\times$ (level 2)
Sommer et al. (2020)	2 neuroradiologists, blinded, 28 clinical cases with real motion	Artifact severity (5-point scale) before vs after network correction	Significant reduction in mean artifact scores ($p < .03$); reference-based metrics inapplicable, as the reference itself is motion-corrupted

$$\begin{aligned}\mathrm{MSE}(\hat{X}, X) = \frac{V_{\text{lesion}}}{V_{\text{total}}}\mathrm{MSE}_{\text{lesion}} \\ + \left(1 - \frac{V_{\text{lesion}}}{V_{\text{total}}}\right)\mathrm{MSE}_{\text{bg}}.\end{aligned} \tag{1}$$

Table 6: Evaluation practice by the highest acceleration factor (R) each study reports, among the 63 of 263 included studies that state one. Bands are half-open, so a study reporting exactly $R = 4$ falls in the second band and one reporting exactly $R = 8$ in the third. Counts are studies; percentages are within band. *Reader*, any observer or reader assessment recorded; *M+R*, a pixel-wise fidelity metric and a reader assessment recorded on the same data; *Halluc.*, hallucination or fidelity instability among the recorded failure modes; *Artifacts*, aliasing, ghosting or Gibbs artifacts among them. Acceleration factors were extracted from full text after the registered extraction round, each with a verbatim supporting quotation and excluding matches inside reference lists and citation contexts; the 407 retained quotations come from the 63 studies that state a factor, a mean of 6.5 quotations per such study, and are released as `acceleration_factors_evidence.csv`. The extraction is screening-level, so counts are lower bounds. No study at any acceleration factor reports a rank correlation between a fidelity metric and reader judgement.

Acc. (R)	**Studies**	**Reader**	**M+R**	**Halluc.**	**Artifacts**	**Sequences most often evaluated**
$2 \leq R < 4$	32	7 (21.9%)	1 (3.1%)	7 (21.9%)	25 (78.1%)	T1w 15, T2w 12, FLAIR 11
$4 \leq R < 8$	17	5 (29.4%)	1 (5.9%)	2 (11.8%)	12 (70.6%)	T1w 9, T2w 9, FLAIR 7
$R \geq 8$	14	5 (35.7%)	4 (28.6%)	2 (14.3%)	8 (57.1%)	T1w 9, T2w 8, FLAIR 8
All	**63**	**17 (27.0%)**	**6 (9.5%)**	**11 (17.5%)**	**45 (71.4%)**	T1w 33, T2w 29, FLAIR 26

The R-reporting subset is close to the whole corpus on reader assessment (27.0% against 21.7% for all 263), so it is not a systematically better-evaluated stratum. Reader assessment rises with acceleration (21.9% → 29.4% → 35.7%) and paired metric–reader reporting concentrates in the most accelerated band (4/14 against 1/32 in the $2 \leq R < 4$ band), but the counts are small and most studies in every band (including 9 of the 14 with $R \geq 8$) report no reader assessment at all. Artifact description is the most frequently recorded failure mode at every acceleration level, whereas hallucination or fidelity instability is recorded in 11 of 63.

With lesion volume fraction $f = V_{\text{lesion}}/V_{\text{total}}$ and local error ratio $r = \text{MSE}_{\text{lesion}}/\text{MSE}_{\text{bg}}$, the global PSNR change caused by a localized edit follows directly:

$$|\Delta\text{PSNR}| = 10 \log_{10}\left(1 + f(r-1)\right) \quad (\text{dB}). \tag{2}$$

The consequence is quantitative rather than rhetorical, and it is worth anchoring on a lesion a neuroradiologist would expect to resolve rather than on the smallest arithmetically convenient one. An acute lacunar infarct of 100 mm^3 is a sphere 5.8 mm across: on a clinical DWI acquired at 4–5 mm section thickness (Lang et al., 2024) that is several voxels wide in-plane and spans one to two sections, so it is resolved, if only just, in the through-plane direction. It occupies a volume fraction $f = 6.3$–8.3×10^{-5} of a 1.2–1.6 L brain. Even at a hundredfold local

Table 7: The four mechanisms by which a whole-volume fidelity score fails to register a focal diagnostic error (RQ2). The third column names a failure the mechanism permits, not one this review has observed at a stated frequency; the fourth states why the score does not move. Numerical entries are computed from Equation (2) under the stated error model, not measured in any primary study.

Mechanism	**Formal statement**	**Clinical failure it permits**	**Why the score does not move**
(a) Spatial error averaging	L_2 error is averaged uniformly over all $H \times W \times D$ voxels, so a lesion contributes with weight $f = V_{\text{lesion}}/V_{\text{total}}$	Erasure of an acute lacunar infarct on DWI or FLAIR, or of a punctate microhaemorrhage on SWI or T_2^*	By (2), erasing a 100 mm^3 infarct at $r = 100$ moves global PSNR by 0.027–0.036 dB; a 5 mm^3 microhaemorrhage by $1.3\text{–}1.8 \times 10^{-3}$ dB
(b) Reference dependence	Every reference-based score presupposes a registered, artifact-free reference image	A valid motion-corrected reconstruction is penalised against a corrupted reference, and genuine correction is scored as error	The comparison is no longer between reconstruction and truth; PSNR and SSIM become uninterpretable rather than merely noisy
(c) Loss symmetry	$(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$: the objective is indifferent to the sign of the error	A suppressed hyperintense lesion is scored identically to added background noise of the same magnitude	Diagnosis is asymmetric where the metric is not: a missed finding and a spurious brightening carry the same penalty
(d) Realism–reward trade-off	Adversarial and perceptual objectives reward outputs typical of the training distribution, $\log D_\phi(\hat{X})$, not outputs faithful to the measurement	Synthesis of plausible vessels, sulci or texture that was never acquired	Perceptual scores can rise as measurement fidelity falls, so the objective pays for the hallucination it should penalise

error ratio ($r = 100$), Equation (2) gives $|\Delta\text{PSNR}| = 0.027\text{–}0.036$ dB. Erasing a resolvable acute infarct outright therefore moves global PSNR by about three hundredths of a decibel, which is the same order as the third decimal place of a score conventionally reported to two.

The limiting case is worth stating precisely, because it is easy to state loosely. A punctate cerebral microhaemorrhage of 5 mm^3 is 2.1 mm across. It is *not* resolved on clinical DWI, whose voxels are roughly 12–20 mm^3, so no reconstruction can be said to erase it there; it is not resolved in the first place. It is several voxels across on the high-resolution SWI or T_2^*-weighted acquisition on which microhaemorrhages are actually read, and on that sequence its era-

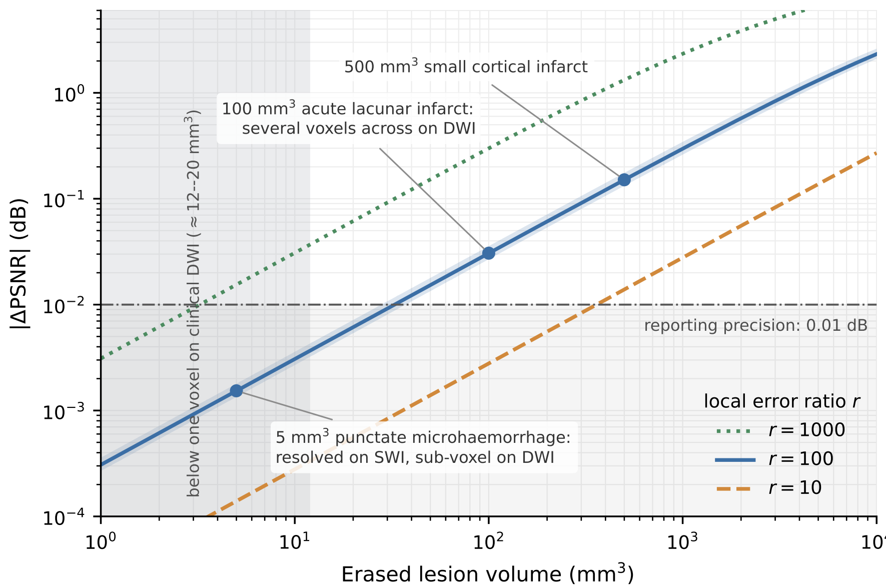

Figure 2: The PSNR cost of erasing a focal lesion, from Equation (2). Curves give $|\Delta\mathrm{PSNR}|$ against erased lesion volume for three local error ratios r at a 1.4 L brain; the shaded band around $r = 100$ spans the 1.2–1.6 L range. Markers show the three clinical anchors used in the text. The horizontal line at 0.01 dB is the two-decimal precision at which PSNR is conventionally reported, and the shaded left-hand region is the volume below a single voxel on a clinical DWI acquisition, where erasure is not defined because the finding is not resolved. Every value is computed from the identity under the stated error model; nothing in this figure is measured.

sure costs $f = 3.1\text{–}4.2 \times 10^{-6}$ and $|\Delta\mathrm{PSNR}| = 1.3\text{–}1.8 \times 10^{-3}$ dB, below the precision at which the score is reported at all. Scaling in the other direction, a 500 mm^3 cortical infarct costs ≈ 0.15 dB (0.13–0.18 dB across the same brain-volume range). Fig. 2 plots the identity across this range, with the sub-voxel region marked. The argument does not need the extreme case: it holds by one to two orders of magnitude at a lesion size that is unambiguously resolved and unambiguously matters. These are computed bounds from the stated f and r, not measurements from any primary study, and the error model behind them, a uniform background error and a free local error ratio, is an assumption we state rather than a property of any reconstruction.

Loss-function symmetry. MSE and SSIM are indifferent to the sign of the error, since $(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$, whereas diagnosis is not: suppressing a true hyperintense lesion means the finding is missed, while adding background noise costs nothing clinically. An evaluation respecting that asymmetry must penalise the two directions differently, for example

$$\mathcal{L}_{\text{asym}}(\hat{X}, X) = \alpha \sum_{i\in\Omega_{\text{lesion}}} \max(0, X_i - \hat{X}_i)^2 + \beta \sum_{i\in\Omega_{\text{bg}}} (\hat{X}_i - X_i)^2, \quad (\alpha \gg \beta). \tag{3}$$

Several included studies already build objectives of this shape: lesion-mask-weighted losses motivated explicitly by the dilution of lesion gradients among background voxels (Lee et al., 2026a), overlap-weighted reconstruction losses that up-weight structure against background (Jayakumar et al., 2023), and gradient-based structure-preservation terms (Zhang et al., 2026). Their existence is evidence that the symmetry problem is recognised in practice, though none of these studies pairs the correction with a reader assessment.

Realism–reward trade-off. Adversarial and perceptual objectives reward outputs that look like the training distribution, which is not the same as outputs faithful to the measurement, so a generative reconstruction can gain plausible texture that was never acquired (Zhou et al., 2024; Bhadra et al., 2021).

Reference dependence. All four metrics presuppose a registered, artifact-free reference. Intra-scan motion invalidates that assumption, and 69 of 263 studies rest their reference standard solely on a fully sampled acquisition, which is valid only where the subject did not move.

4.4. RQ3: what current evaluation paradigms can and cannot establish

Read against those mechanisms, the corpus supports a paradigm-by-paradigm account of what each evaluation approach can demonstrate about prior-induced diagnostic error. Table 8 states the account compactly; the argument behind it is given here, because two of its six rows are analytical claims we derive rather than findings we extracted, and an analysis presented only in a table has not been argued.

What the four empirical paradigms establish. Pixel-wise fidelity metrics establish aggregate agreement with a reference and nothing about focal change:

by (2) the penalty for erasing a resolvable acute infarct is of the order of the reporting precision, so a method can pass any threshold anyone would set while failing on the finding that matters. Perceptual and rank-trained metrics track reader preference better than MSE or SSIM as an optimisation target, but they inherit the variability of the readers they were trained on and, in the one study of this kind in the corpus, were calibrated on synthetically corrupted images rather than on reconstructions of undersampled k-space (Tang et al., 2025), so their transfer to the reconstruction setting is assumed rather than shown. Uncertainty estimates flag where a model is unsure, which is not the same question: a prior confident enough to remove a lesion is confident in the region where it removed it, so posterior variance can be low exactly where the diagnostic error is. Task-based observer assessment does settle the question directly, and it is the only paradigm here that does, which is why its scarcity in this corpus (57 of 263 studies, and 18 paired with a metric) is the review's central finding rather than an incidental one.

Two limits that follow from the physics, not from the corpus. The first concerns the check that reconstruction papers most often use to argue that an output is faithful to the data: the measurement-space residual $\|\mathbf{A}\hat{X}-Y\|_2/\|Y\|_2$ against a tolerance. This check cannot detect prior-induced edits at all, and the reason is structural rather than statistical. Write the reconstruction as the sum of its measured and unmeasured components, $\hat{X} = \mathbf{A}^+\mathbf{A}\hat{X} + (\mathbf{I}-\mathbf{A}^+\mathbf{A})\hat{X}$. The second term lies in the null space $\mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ annihilates it by definition, so any amount of structure written there, an erased lesion or an invented vessel alike, leaves the residual exactly unchanged. Undersampling is what creates that null space; the more aggressive the acceleration, the larger the subspace in which a learned prior can write freely without moving the number the field uses to certify data consistency. A residual check is therefore a two-sided measurement-fidelity band and not a safety test, and its apparent reassurance grows least trustworthy precisely where acceleration is highest.

The second limit is the constructive corollary. If prior-supplied content lives in $\mathcal{N}(\mathbf{A})$, then isolating $(\mathbf{I}-\mathbf{A}^+\mathbf{A})\hat{X}$ and inspecting it is the natural place to look for erasure and hallucination, since that is where both must reside (Bhadra et al., 2021; Gottschling et al., 2025). This is the one paradigm in Table 8 that is not represented in the corpus at all, and its practical obstacle is the same one that limits third-party auditing generally: computing the projection requires the forward operator, hence the raw k-space and the coil sensitivities, which commercial inline pipelines do not export (Section 5). Causal and counterfactual auditing, which asks the most direct question of all, whether a specific finding survives a controlled change of acquisition, appears in a small part of the corpus and almost never for physics-constrained k-space reconstruction (Pawlowski et al., 2020; Güngör et al., 2023; Peng et al., 2026).

The answer to RQ3 is therefore not that current evaluation is weak but that it is mismatched: each paradigm in use answers a question adjacent to the safety question, and no paradigm now in use establishes reconstruction safety on its own. The two that could, task-based observer assessment and null-space inspection, are respectively rare and absent, and the requirements in Section 4.7 follow from that pairing.

4.5. Structure of the evaluation gap

The counts above establish that safety-relevant evaluation is scarce. Analysis of the same per-study records establishes where the scarcity comes from, and indicates a division of labour rather than uniform neglect.

The benchmarks cannot show the failure the field is worried about. One hundred and nine of the 263 studies (41.4%) name a public dataset, and across the whole corpus those references resolve to eleven datasets: fastMRI, the Human Connectome Project, ADNI, IXI, UK Biobank, BraTS, OASIS, Calgary–Campinas, M4Raw, the developing Human Connectome Project and TCGA. Dataset labels are multi-valued (136 cohort assignments across the 109 studies), and of the 109, the cohorts named are healthy volunteers or population samples (55, 50.5%), chronic neurodegenerative cohorts (31, 28.4%), brain tumour cohorts (16, 14.7%), or raw-k-space benchmarks not curated for pathology (34, 31.2%). These four shares are multi-label and sum to 124.8%, a study naming

Table 8: What each evaluation paradigm can and cannot establish about prior-induced diagnostic error (RQ3). Rows 1–4 summarize evaluation practice and performance as reported in the included studies; quoted values are representative results from the cited primary studies, not pooled estimates. Rows 5–6 state analytical limits derived by the authors from the acquisition physics (our own analysis; Section 4) and are labelled as such rather than attributed to any primary study.

Evaluation Paradigm	**Core Mechanism**	**What It Can and Cannot Establish**	**Key References**
1. Pixel-wise fidelity metrics	L_2 / SSIM whole-image voxel averaging	Aggregate agreement with a reference only. Cannot resolve focal change: by (2), erasing a resolvable 100 mm^3 lacunar infarct at $r = 100$ costs 0.03 dB. Requires a registered reference	Recorded in 84 of the 144 evaluation-labelled studies
2. Perceptual & learned metrics	Networks trained on radiologist rankings	Tracks reader preference better than MSE/SSIM as an optimisation target ($75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ against readers' own $61.5\% \pm 5.5\%$). Inherits training-reader variability, and was calibrated on synthetically corrupted data rather than clinically undersampled acquisitions, so transfer is assumed (Section 4.4)	Tang et al. (2025) ($n = 18$ in corpus)
3. Uncertainty estimation	Posterior sampling / Bayesian variance $p(X \mid Y)$	Flags out-of-distribution inputs and regions of low confidence. Does not attribute missing structure to prior vs sampling operator; high confidence in hallucinated regions is possible	Khawaled and Freiman (2024) ($n = 18$ in corpus)
4. Task-based observer assessment	Human reader performance on a diagnostic task; model observers are its automated form, unrepresented here (Barrett et al., 1993; Barrett and Myers, 2004; International Commission on Radiation Units and Measurements, 1996)	Establishes diagnostic sufficiency directly; the gold standard in assessment theory. Costly and reader-variable ($61.5\% \pm 5.5\%$, falling to 25% under absolute Likert scoring), and rarely paired with a metric: only 18/263 report both	Radmanesh et al. (2022), Lang et al. (2024) ($n = 57$ in corpus)
5. Measurement-space residual check	$\lVert\mathbf{A}\hat{X} - Y\rVert_2 / \lVert Y\rVert_2$ against tolerance	**Cannot detect prior-induced edits**: the residual is invariant to anything in $\mathcal{N}(\mathbf{A})$. A two-sided measurement-fidelity band, not a safety test	Common diagnostic check in the reviewed studies
6. Null-space / hallucination mapping	Structure written into the null-space component $(\mathbf{I} - \mathbf{A}^{+}\mathbf{A})\hat{X}$	Isolates prior-supplied content, where erasure and hallucination must reside. Requires $\mathbf{A}$, hence raw k-space and coil sensitivities	Bhadra et al. (2021), Gottschling et al. (2025); formalization for reconstruction auditing is author-derived

two cohorts being counted in both; every other percentage here is single-label. *Within this corpus, no study benchmarks against a dataset curated for acute ischemic stroke or intracranial haemorrhage.* The sharper version of this finding sits in the annotation layer the field has already built. fastMRI+ supplies 7,570 expert bounding-box annotations across thirty brain pathology categories (Zhao et al., 2022), and not one of the thirty is a haemorrhage category of any kind, microhaemorrhage included. The apparent counterexample should be stated plainly rather than left for a reader to find, because it is the category closest to this article's own worked example: fastMRI+ *does* carry an image-level *Lacunar Infarct* label (113 annotations across 32 subjects), alongside study-level *Small Vessel Chronic White Matter Ischemic Change* and a *Global Ischemia* label recorded for a single subject. It does not rescue the benchmark, and the reasons are the ones that matter for reconstruction safety. No fastMRI+ category distinguishes acute from chronic infarction, so a lacunar label carries no acuity and cannot identify the acute presentation whose erasure is at issue. The limit is in the source images as much as in the label set: the annotated series are axial T2-FLAIR, T1w or contrast-enhanced T1w only, so neither DWI, on which infarct acuity is established, nor SWI or T_2^*-weighted imaging, on which microhaemorrhage is read, was annotated at all. Acute vascular pathology is therefore absent even where annotation infrastructure exists, and it is absent for a reason that adding labels alone would not repair. Supplementary Table S10 sets out the full derivation, including the sub-selection of 1,001 of the 5,847 fastMRI brain studies, the category listing, and the single-annotator limitation that travels with these annotations. Stroke-lesion datasets do exist, ISLES (Hernandez Petzsche et al., 2022) and ATLAS (Liew et al., 2022) among them, but they are segmentation benchmarks distributed as processed images without raw k-space, so they cannot serve reconstruction evaluation as released. This matters mechanically: lesion erasure cannot be measured on cohorts that contain no acute lesions, so a model can lead every public leaderboard in the field while remaining untested against the failure mode that would matter most in an emergency department. Where a pathology-labelled dataset is used at all it is BraTS, whose tumours are large, high-contrast mass lesions, close to the easiest possible case for lesion preservation and the least informative about the lacunar and punctate findings at issue here. The benchmark infrastructure and the metric therefore fail in the same direction.

Rigour is divided between two literatures. Of the 263 studies, 84 (31.9%) release code but report no reader assessment, 44 (16.7%) report a reader assessment but release no code, 122 (46.4%) do neither, and only 13 (4.9%) do both. The division tracks publication venue. Applying the post-hoc classification rule stated in Section 3 to all 263 venues, clinical imaging journals ($n = 131$) carry reader assessment at nearly three times the rate of engineering and methods venues ($n = 78$), which in turn release code at nearly twice the clinical rate; general neuroscience and neuroimaging venues ($n = 37$) sit between. Supplementary Table S12 gives the full cross-tabulation. The rule leaves 17 of the 263 journal titles unmatched, and those are reported as a fourth *unclassified* group rather than forced into one of the three; the four groups between them account for all 57 reader studies and all 97 code releases. The tendency for the two practices to fall in different studies rather than the same ones gives a Fisher exact odds ratio of 0.43 ($p = 0.013$). The estimate is insensitive to the one labelling choice that could move it: taking the reader subset as the 61 studies originally labelled at extraction rather than the 57 verified at appraisal gives an odds ratio of 0.43 and $p = 0.010$. We nonetheless report the association descriptively rather than as a hypothesis test, because the venue grouping it rests on is a post-hoc keyword rule that was not part of the fixed protocol, and a p-value computed on a partition chosen after seeing the data does not carry its nominal meaning. The direction and magnitude are what we rely on. The magnitude is the part that does not depend on the choice: computational verifiability and diagnostic verifiability are largely held by different communities, only one study in twenty holds both, and a safety argument requires both.

Awareness of a failure mode does not produce the evaluation that would detect it. Generative studies record hallucination or fidelity instability in 31/80 (38.8%) of cases, the highest share among the four largest method families, against

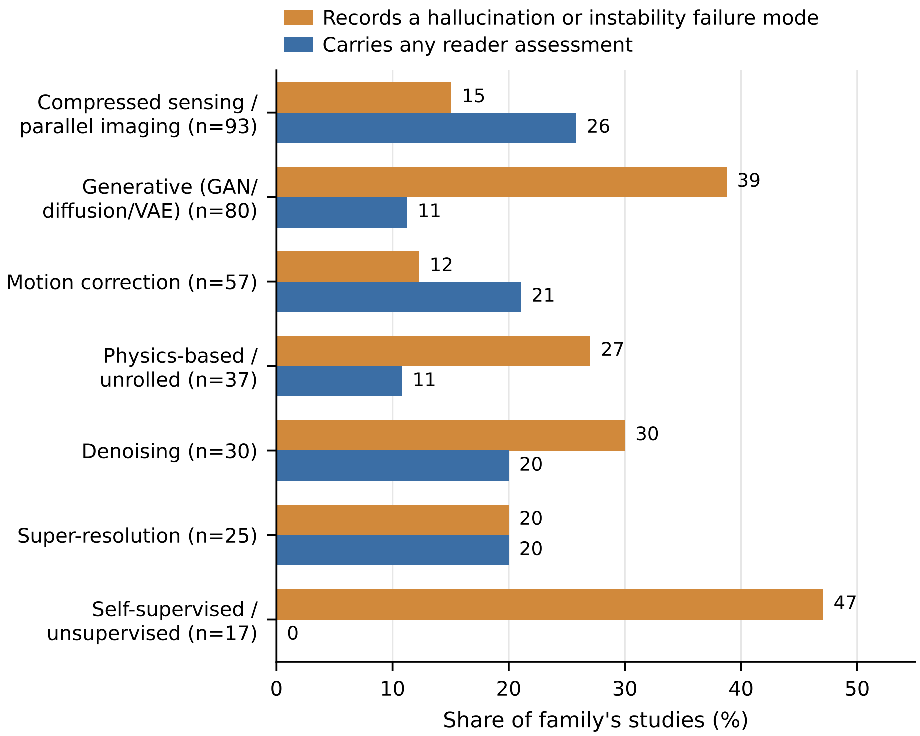

Figure 3: Reader assessment against hallucination awareness, by method family (recorded extraction labels; lower bounds). Bars are the share of each family's studies that record a hallucination or instability failure mode (orange) and the share that carry any reader assessment (blue), using the verified reader subset ($n = 57$). Bar labels are rounded to the nearest integer; the one-decimal values quoted in the text are the same quantities (for example 15 against 15.1%, 39 against 38.8%). Generated from the released per-study file (`included_-characteristics.csv`) by `5_Code/make_figures.py`; every value is recomputable from the deposit.

15.1% for compressed-sensing and parallel-imaging work and 12.3% for motion correction. Yet at 9/80 (11.3%) they are also among the two least reader-evaluated of those families, while the small self-supervised group records the corpus's highest hallucination share (8/17, 47.1%) and no reader assessment at all (0/17); per-family shares are in Table 9 and Fig. 3. The sub-fields that most clearly recognise that their models can synthesise structure are the least likely to run the test capable of detecting it.

Evaluation is not broadening as the field grows. Reader-assessment share fell from 32.4% (1995–2019) to 18.2% (2020–2022), recovering only partially to

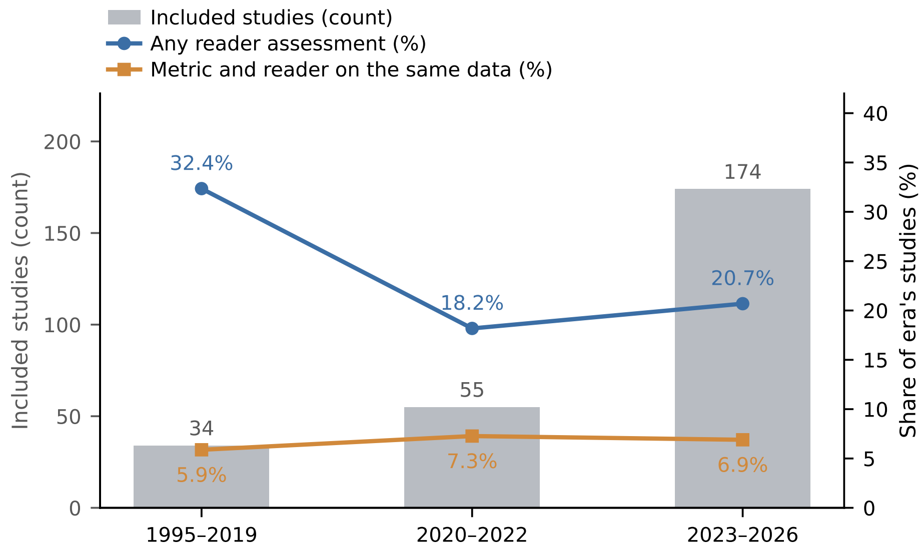

Figure 4: Evaluation practice across publication eras. Grey bars give the number of included studies per era (34, 55, 174); lines give the share with any reader assessment and the share co-reporting a fidelity metric and a reader assessment on the same data, using the verified reader subset ($n = 57$). What recovery there is in 2023–2026 comes from subscription-walled clinical journals (52.0% reader assessment in that stratum against 15.4% in the open remainder; Section 5.2). Generated from the released per-study file by `5_Code/make_figures.py`; counts are lower bounds from recorded extraction labels.

20.7% (2023–2026) while the corpus grew fivefold (Table 9 and Fig. 4). What recovery there is comes entirely from the subscription-walled clinical literature: among 2023–2026 studies obtainable only through title-level library subscriptions, reader assessment runs at 52.0% (13 of 25), against 15.4% (23 of 149) in the openly accessible remainder of that era. Reader evidence, where it exists, sits disproportionately behind paywalls, and a review that cannot retrieve those texts will understate it; Section 5.2 takes the direction of that bias seriously rather than dismissing it. Among the 144 studies recording any evaluation paradigm, four in five record exactly one, and the share recording two or more fell across the same periods (Supplementary Table S12). Of the 84 studies recording a pixel-wise metric, 60 (71.4%) record no reader, uncertainty or perceptual measure beside it.

Readers are not displaced away from the sequences where the stakes are highest. One expectation is not borne out. The comparison must be made

Table 9: Evaluation practice across publication eras and method families (recorded labels; lower bounds). Reader-assessment share fell as the corpus grew and recovers only partially in the latest era, and that recovery is carried by subscription-walled clinical journals. Among the four largest method families the generative group, the one most associated with hallucination, is one of the two least reader-evaluated and is indistinguishable from physics-based unrolled work; the small self-supervised family ($n = 17$) records no reader assessment at all. Reader shares use the verified reader subset ($n = 57$; Section 3), so counts sum to 57 across the era rows and are multi-label across the family rows.

Era	**n**	**Reader**	**Uncert.**	**Co-report**
1995–2019	34	32.4% (11/34)	5.9%	5.9%
2020–2022	55	18.2% (10/55)	5.5%	7.3%
2023–2026	174	20.7% (36/174)	7.5%	6.9%

Method family (multi-label)

Family	**n**	**Reader share**
Compressed sensing / parallel imaging	93	25.8% (24/93)
Motion correction	57	21.1% (12/57)
Super-resolution	25	20.0% (5/25)
Denoising	30	20.0% (6/30)
Generative (GAN/diffusion/VAE)	80	11.3% (9/80)
Physics-based / unrolled	37	10.8% (4/37)
Self-supervised / unsupervised	17	0.0% (0/17)

on disjoint strata to mean anything, because pulse sequence is a multi-label field and the naive contrast double-counts the 107 studies that acquire both an acute-pathology and a structural sequence (Supplementary Table S11 gives the correction). That the two disjoint strata below sum to $23 + 84 = 107$, the same number, is a third coincidence of unrelated tallies rather than a transcription error, verified against the released per-study file as the two flagged in Section efsec:requirements are. Restricted to studies acquiring one group and not the other, reader assessment runs at 5/23 (21.7%) for DWI or FLAIR without T1w or T2w against 10/84 (11.9%) for T1w or T2w without DWI or FLAIR — a difference in the direction one would hope for, resting on 23 studies in the smaller stratum and not approaching significance (Fisher exact odds ratio 2.06, $p = 0.31$). We report it as descriptive and draw only the negative conclusion it supports: there is no evidence that reader assessment is *displaced away* from the sequences on which acute pathology is read. The deficit is one of scarcity and of distribution across research communities, not of misdirection within the clinical literature.

4.6. Methodological quality and risk of bias

Across all 57 QUADAS-2 appraised diagnostic and reader studies, methodological quality was highest for Flow and Timing (45/57 low risk of bias, 78.9%), reflecting the standard retrospective undersampling paradigm where reconstruction and reference share the same acquisition timepoint without temporal degradation. Reference Standard and Index Test both sit at 26/57 low risk (45.6%), each now below half, with high ratings arising where radiologists scored accelerated images with reference ground truth visible or where lesion truth was defined by the comparator images themselves (incorporation bias). Patient Selection remains the weakest domain (15/57 low risk, 26.3%; 19/57 unclear, 33.3%; 23/57 high risk, 40.4%), as cohorts were predominantly retrospective, single-centre, and pathology-enriched without explicit prospective enrolment rules. Applicability concerns follow a similar distribution, with low concerns in Patient Selection for 36/57 (63.2%), Index Test for 30/57 (52.6%), and Reference Standard for 39/57 (68.4%).

The two rounds do not place the weaknesses in quite the same order, and the pooled shares above differ from the first round's in every one of the seven domains. Low-risk shares fell for Index Test (53.8% to 45.6%), applicability of the Index Test (59.0% to 52.6%), Flow and Timing (82.1% to 78.9%), Reference Standard (46.2% to 45.6%) and applicability of Patient Selection (64.1% to 63.2%); they rose for Patient Selection risk of bias (23.1% to 26.3%) and applicability of the Reference Standard (59.0% to 68.4%). The two largest movements concern the index test and the applicability of the reference standard, and the first of these runs against the appraised studies. The second-round studies are therefore not a favourable addition to the appraisal, which is why the pooled figures are the ones reported here. The first-round shares quoted here already incorporate the five studies re-appraised on recovered full text.

Corpus-wide reporting transparency evaluated against STARD principles revealed substantial omissions across the broader 263 studies: 109/263 (41.4%) provided no identifiable reference standard (47 stating none at all, Table 3, and a further 62 naming a ground truth they left unspecified), scanner field strength was unreported in 54/263 (20.5%), and 166/263 (63.1%) did not make code openly available. Overall, 220/263 studies (83.7%) omitted at least one key methodological indicator (field strength, reference standard, or open code). Two directions must be kept apart here, and conflating them has caused confusion elsewhere in this literature. With respect to the *reconstruction methods*, this reporting and selection pattern is anti-conservative: retrospective single-centre evaluation on curated data flatters reported performance relative to what prospective clinical deployment would show. With respect to *this review's own conclusion*, the same pattern is conservative: if published performance is flattered and failure is under-reported, then the scarcity of safety-relevant evaluation we document is an understatement, and the true gap is at least as large as the one measured. The safety argument turns on the second direction and is not weakened by the first. Detailed domain-level judgments and supporting quotations are available in the open project repository.

4.7. Requirements for safety-oriented evaluation

Read together, the synthesis implies five requirements that any evaluation intended to establish reconstruction safety must meet, each stated with the corpus evidence that motivates it and none met by more than a small minority of included studies. **R1, focal task-based measurement**: every global fidelity score must be accompanied by a region-of-interest or task-based observer measure, because spatial averaging is the first mechanism by which lesion-scale error disappears (60/84, or 71.4%, of pixel-metric studies record nothing beside the metric; this equals the artifact share of the R-reporting subset in Table 6 (45/63) to the decimal, which is a coincidence of two unrelated tallies rather than a transcription error, verified against the released per-study file). **R2, paired metric–reader reporting**: metric validity claims require a reader comparison on the same data (6.8% of studies provide one). **R3, reference-independent validity**: evaluation must remain interpretable when no registered artifact-free reference exists (26.2% rest solely on a fully sampled acquisition). **R4, robustness under acquisition shift as a primary outcome**, reported as a result rather than a caveat; robustness or distribution shift is recorded as a failure-mode concern in 76/263 studies (28.9%), almost always in the discussion rather than as a pre-specified endpoint. **R5, hallucination-specific detection**, operating on the image side, since measurement-space residual checks are provably blind to null-space edits. To these the analysis in Section 4.5 adds a precondition: benchmarks containing the pathology whose erasure is at issue, without which R1, R2 and R5 cannot be exercised at all.

5. Discussion

Three conclusions follow. First, the comparison on which the field's central proxy rests is almost never made: no included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment, and only 18 of 263 record the two on the same data at all. Where readers do appear, what is measured is mostly agreement between readers, and it is weak and condition-dependent. The finding therefore rests more on the absence of measurement than on the measurements that exist. Second, the reason is structural, through the four mechanisms of Section 4: background voxels average focal error away, the loss functions are sign-symmetric where diagnosis is not, adversarial terms pay for invented texture, and every one of these metrics assumes a registered reference that motion destroys. The mechanisms are mathematical properties of the metrics themselves; how often each matters clinically is the part the thin reader evidence leaves less certain. Third, no evaluation paradigm now in use settles the safety question by itself, a conclusion resting on the paradigm-by-paradigm synthesis of Section 4.

The analysis of where the gap comes from, which was not part of the fixed protocol and is reported as exploratory throughout, changes what would follow from these findings if it holds. The shortfall does not look like a uniform oversight; it co-occurs with two structural features. Evaluation practice is divided between a community that releases code and a community that runs readers, with only 4.9% of studies doing both (an association we interpret descriptively in Section 4.5, its venue partition being post-hoc), and the benchmark datasets named in this corpus contain no acute pathology, so the failure mode of greatest clinical consequence is not merely under-measured but unmeasurable on the datasets in common use. If that association is causal, which this design cannot establish, recommendations addressed to individual authors will not close the gap; what would be required is benchmark construction and reporting conventions that force the two halves of the evidence together.

This matters because accelerated reconstruction has left the research setting. Deep-learning acceleration, denoising and image enhancement ship inside commercial MRI products, several of them cleared as software as a medical device: one such tool, an FDA-cleared vendor-agnostic DICOM-domain enhancement product, is evaluated in this corpus by a prospective multicentre multireader trial (Bash et al., 2021), which is also a reminder that a cleared product can be validated on one sequence family and deployed across others. Published guidance in both jurisdictions increasingly asks manufacturers to demonstrate performance for the device's stated intended use and under conditions representative of deployment rather than on curated development data: the FDA, Health Canada and MHRA guiding principles on good machine learning practice ask that datasets be representative of the intended patient population and that testing be independent of training (U.S. Food and Drug Administration et al., 2021), the EU AI Act imposes accuracy and robustness obligations on high-risk AI systems (European Parliament and Council of the European Union, 2024), and the Medical Device Regulation requires clinical evidence appropriate to the device's intended purpose (European Parliament and Council of the European Union, 2017). Our reading of that guidance is that aggregate pixel-wise fidelity cannot support a claim about small-lesion detection, because (2) shows the aggregate does not move when the small lesion is removed. That reading is ours: no regulator has reviewed or endorsed it, and none of the three instruments names an image-quality metric.

A practical obstacle compounds it, and it is the reason the requirements in Section 4.7 are addressed to infrastructure rather than to authors. Commercial inline pipelines export DICOM images after undisclosed normalisation, coil combination and vendor post-processing, and the raw complex k-space typically never leaves the scanner. Every method in Table 8 that could detect a prior-induced edit needs something the exported DICOM does not carry: the null-space projection needs the forward operator and the coil sensitivities, a counterfactual audit needs the ability to re-run the acquisition under a controlled change, and even a measurement-space residual needs Y. Independent third-party auditing of a deployed reconstruction is therefore not merely difficult but structurally impossible without an open raw-data interface, and no amount of reporting discipline by authors of research papers can supply one.

Three things follow that are worth stating as claims rather than as recommendations.

The cheapest useful change costs no new data. A reconstruction paper that claims its method preserves small lesions can state, from (2) and its own brain volume and assumed local error ratio, the change in global PSNR that erasing the smallest lesion it claims to preserve would produce. For a lacunar infarct that is about 0.03 dB, which is smaller than the difference the paper is likely to be reporting between its method and its baseline. This is arithmetic, not a new experiment, and it converts an implicit claim into an explicit one at the point where the claim is made. Nothing in this review suggests such a line would be difficult; the finding is that nobody currently reports it.

The two-community split is self-reinforcing, which is why exhortation has not worked. Each community's evaluation is close to the best it can do with the data it can obtain. Engineering venues evaluate on the datasets that carry raw k-space, and those contain healthy volunteers, chronic cohorts and tumours rather than acute pathology, so a reader study run on them would not test lesion preservation even if it were run. Clinical venues have the pathology, the readers and the acute presentations, and no access to raw k-space or to the vendor's reconstruction, so a reproducibility claim is not available to them. The 4.9% of studies doing both are not more diligent than the rest; they are the studies that happened to sit where both kinds of access coincide. An instruction to individual authors to run more reader studies asks most of them to do something the data available to them makes uninformative.

The missing object is a dataset, and its requirements are now specifiable. The public resources discussed in this review supply each necessary component separately, and the distinction between what the corpus uses and what merely exists is part of the point: raw multi-coil k-space is available in three of the eleven datasets the included studies actually name (fastMRI, Calgary–Campinas, M4Raw); acute cerebrovascular pathology with expert segmentation exists in ISLES and ATLAS, which no included study uses because they are distributed as processed images; and reader-level pathology annotation on brain MRI exists in fastMRI+, which no included study uses either. None combines them, and the combination is what every requirement in Section 4.7 presupposes: raw k-space so the forward operator is known, acute lesions confirmed against a standard independent of the images being evaluated, lesion-level annotation so a focal figure of merit can be computed, and a licence permitting redistribution so the benchmark can be used by people who did not collect it. That object does not exist, and until it does the safety question is not merely unanswered but unaskable on public data.

5.1. Priorities for future research

Three of the five requirements in Section 4.7 translate directly into research priorities, and we do not restate them here: paired metric–reader reporting (R2), robustness under acquisition shift reported as a primary endpoint rather than a caveat (R4), and benchmarks carrying the pathology whose erasure is at issue, which is the precondition on which R1, R2 and R5 all depend. Calibrated voxel-level uncertainty that tracks local reconstruction error (Khawaled and Freiman, 2024) belongs with R4 and is the form in which R4 is most likely to be actionable for generative and self-supervised architectures.

A fourth priority does not follow from anything this review extracted, and we separate it for that reason. Many current pipelines operate slice by slice and introduce inter-slice discontinuities on isotropic 3D acquisitions, so reconstruction and auditing frameworks should model 3D anatomical coherence directly (Peng et al., 2026) while meeting scanner-side latency constraints. Slice-wise versus volumetric processing was not an extraction field in this review. We state the priority on mechanistic grounds and on the cited work, not on a corpus count.

5.2. Threats to validity

The strongest conclusions rest on a small subset of reader-based studies, and we state that prominently rather than in passing: 57 of 263 studies carry any reader assessment and 18 pair it with a metric. Four threat classes bound what follows from that base.

How to read the limitations below. They are not all bounded in the same way, and treating them as one undifferentiated list of caveats would misrepresent what this review knows about itself. Three kinds appear. Some are *bounded by released evidence*: the non-retrieval analysis, the paywall stratification and the appraisal-agreement decomposition are each quantified from files a reader can recompute. Some are *bounded by argument*: the direction of publication bias and of the reporting asymmetry is reasoned rather than measured, and we say which way each points. One is simply *unbounded*, and it is the sharpest: extraction has no reliability estimate.

Three properties of the extraction layer follow from that and should be weighed together. First, its depth is deliberately shallow. Method, failure-mode and evaluation categories were assigned from title, abstract and controlled vocabulary, so the prevalence counts are floors on what studies *report at that level* rather than measurements of what they did. Second, the shallowness is visible in the counts themselves: only 84 of 263 studies are recorded as reporting a pixel-wise fidelity metric, and 39 of the 57 verified reader studies are recorded as reporting none, which for this literature is implausible as a description of practice and entirely plausible as a description of abstracts. A further 119 of 263 studies (45.2%) carry no evaluation-paradigm label at all, which is why we report both the /263 and the /144 denominators wherever the distinction matters. Third, the headline 18/263 mixes two depths: the reader label was verified against full text and corrected at appraisal, while the metric label was not. That asymmetry biases the intersection downward by an unknown factor, and it is the single respect in which our headline figure is least defensible as a point estimate — though the direction is conservative, since correcting it could only widen the measured gap.

What we offer in place of a reliability coefficient is auditability, and the substitution is deliberate rather than a shortfall we failed to notice. Every extracted value is released with the verbatim quotation that supports it, so a reader can check whether an assignment is *correct*, which is a stronger and more useful guarantee than a coefficient showing that two people were *consistent*. It is not a substitute for reliability in the statistical sense, and we do not claim it is: consistency and correctness bound different failure modes, and only the first is estimable from our records. A duplicate-extraction exercise on a random subsample, reported with a κ, is the obvious next step and we name it as an outstanding task rather than a completed one.

Construct validity. We searched with broad, high-sensitivity terms for architectures, artifacts and evaluation paradigms, and the corpus spans the reconstruction-safety construct, from pixel-wise metrics through perceptual losses to reader studies. What the corpus cannot guarantee is that a metric means the same thing wherever it appears: studies differ in what they treat as the quantity of interest, so a rank correlation with reader scores in one paper is not straightforwardly the same measurement as in another, and grouping such values under one heading is an interpretation we make, not a property of the data. The venue classification underlying Section 4.5 is likewise a post-hoc keyword assignment, released with the data rather than extracted from the studies.

Internal validity. A protocol fixed before screening began, a documented search across seven databases, and independent screening and appraisal by three reviewers provide robust protection, and every extracted value is released with the verbatim snippet supporting it. Five specific threats bound these findings. First, 81 of the 1,149 candidate reports remain without obtainable text through structural licensing barriers. Screening the nearest obtainable stratum — the 341 reports recovered only through title-level subscriptions — found 86.2% ineligible on the same criteria, so at that rate the 81 would contribute on the order of 11 studies rather than 81. That bounds the missing evidence without establishing that the bias is nil. Non-retrieval is demonstrably *not* random with respect to the variable this review measures: within the same stratum, 2023–2026 studies carry reader assessment at 52.0% against 15.4% in the openly accessible literature of that era. Both findings point the same way. Reader evidence is over-represented behind paywalls, so a review that cannot retrieve those texts understates how much reader assessment the field performs, and our headline scarcity figures are conservative bounds rather than point estimates. Second, four included studies were assessed on abstract and controlled vocabulary alone and one on partial text; a sensitivity analysis recomputing all headline metrics on the 258 studies verified against complete full text leaves every finding directionally stable (e.g., metric–reader co-reporting remains 7.0% vs 6.8%). Third, every QUADAS-2 tally in this article rests on the whole verified reader subset ($n = 57$, 399 domain judgements), so none is computed on a partial base. The second round moved all seven domain distributions, and the movements that most concern the index test run against the appraised studies: low risk of bias for the Index Test falls from 53.8% to 45.6%, and high applicability concern for the Index Test rises from 33.3% to 40.4% (Section 4). Four studies remain abstract-only and are appraised on that basis. Fourth, reliability is measured for screening and for appraisal but not for extraction, and extraction is the sharpest methodological limitation of the review. The screening and appraisal coefficients, and the decomposition that must be read with the appraisal figure, are given in Section 3 and not restated here; the consequence for this section is that 126 of the 399 QUADAS-2 judgements were reached at a level of agreement a single rater could not be relied on to reproduce, so the domain shares in Table 3 carry that uncertainty. It does not propagate to the prevalence findings, which rest on extraction labels rather than QUADAS-2 ratings, but it bounds how firmly the risk-of-bias picture can be stated. Neither screening nor appraisal agreement transfers to extraction. The duplicate-screening sample is also smaller than the 300–400 records a referee might reasonably ask for, and the interval reflects that. Fifth, four studies recorded at extraction as observer or reader designs were found at appraisal to contain no human reader and were reclassified as algorithmic, moving the reader subset from 61 to 57 and the algorithmic set from 202 to 206. The effect on the headline numbers is small and in the direction of a larger gap: metric–reader co-reporting moves from 7.6% to 6.8%, the 2023–2026 reader share from 23.0% to 20.7%, the code-and-reader intersection from 5.3% to 4.9%, and the Fisher exact p from 0.010 to 0.013 at an odds ratio that moves only from 0.43 to 0.43. Those four corrections are also the source of the 4/61 (6.6%) error rate quoted above, and the design field is the only one that received three-reviewer full-text verification: a comparable rate elsewhere would not overturn any finding here, since the headline shares are separated from their comparators by more than that margin, but it is a source of variability measured in one place and assumed in the rest.

External validity. Among the 209 studies stating a field strength the corpus covers 1.5, 3 and 7 Tesla (21, 153 and 35 respectively), together with several vendors' coil configurations and the major pulse sequences; the remaining 54 (20.5%) state none, and nothing about field-strength coverage should be read into them. Tasks, protocols, architectures, datasets and metrics otherwise vary substantially, which is why we report the direction and consistency of findings rather than their magnitudes. Many algorithmic studies evaluate on retrospectively undersampled single-centre data, which flatters performance relative to prospective multi-centre use; where the public data were themselves preprocessed before undersampling, that inflation is larger again (Shimron et al., 2022). Prospective multicentre reader evidence is thin, and the examples this review leans on each carry a boundary that must travel with them: the emergency-department comparison is single-centre and its one discrepant finding is a single case (Lang et al., 2024); the widest-ranging acceleration reader study is retrospective and single-centre (Radmanesh et al., 2022); and the largest prospective multicentre multireader trial evaluated 3D T1-weighted acquisitions only, with an image-domain DICOM enhancement tool rather than reconstruction from undersampled k-space, reported improved rather than degraded lesion conspicuity, and has several authors affiliated with the vendor of the evaluated tool (Bash et al., 2021) — we cite it for what it does establish, that such tools are cleared and deployed. Finally, the benchmark concentration of Section 4.5 — eleven public datasets named across the corpus, not one curated for acute ischaemic stroke or intracranial haemorrhage — caps how far any leaderboard result can be carried into emergency imaging.

Conclusion validity. Because pixel-wise metrics agree only loosely with radiologist judgement (RQ1), a study resting on PSNR and SSIM alone supports a safety claim less well than one with readers or uncertainty-aware observers, and we weight the evidence accordingly. Publication bias favouring successful reconstructions is a material limitation and works in the same direction, making the reported prevalence of failure evidence an understatement: a literature with little incentive to report failure is exactly the corpus a review about failure must draw on, and the absence of a common effect measure rules out funnel-plot and small-study asymmetry tests. We applied no certainty framework such as GRADE, since the review produces no pooled effect estimates for such a framework to rate. Prevalence counts derive from recorded extraction labels and understate true practice, and no range we quote should be read as an interval estimate rather than as values from named studies.

6. Conclusion

We reviewed 263 primary studies on the safety, artifacts and fidelity of deep learning-based brain MRI reconstruction. Wherever metrics and readers have been compared, agreement between conventional fidelity scores and radiologist judgement is weak, and at most 18 of the 263 studies record the paired metric–reader evaluation capable of exposing that divergence. Every prevalence figure here is a floor derived from screening-level labels rather than a measurement of practice, and although screening agreement was high, the extraction labels those floors rest on carry no reliability estimate of their own. The weakness is structural rather than incidental: the four mechanisms identified under RQ2 explain why, and the safety question is not one that any current evaluation paradigm answers alone.

The contribution we would emphasise is the account of why the gap persists, and we offer it as a hypothesis these data are consistent with rather than as an established structural fact: the venue grouping it rests on is post-hoc, so although the association between code release and reader assessment is strong and would be conventionally significant on a pre-specified partition, we do not read it as a test (Section 4.5). What does not depend on that classification is the magnitude. Only 4.9% of studies both release code and run readers, so computational verifiability and diagnostic verifiability are, as recorded, held by different studies roughly nineteen times in twenty; and no dataset named in this corpus is curated for acute stroke or intracranial haemorrhage. A reconstruction model can therefore satisfy every metric and every public benchmark in current use while the question of whether it preserves a small acute lesion has never been asked. Closing that gap requires benchmarks that contain the pathology at issue and reporting conventions under which no pixel-wise metric appears alone, and until both exist, claims that accelerated reconstruction is diagnostically safe will continue to rest on measurements that cannot see the failures that matter.

One change is available immediately and needs no new data. Equation (2) converts any lesion volume into the PSNR movement its erasure would cause, so an author reporting a reconstruction at a stated acceleration can state, alongside the global score, the smallest focal change that score is capable of registering at their image size. For a 100 mm^3 infarct that figure is around 0.03 dB, which is below the precision at which PSNR is conventionally reported. Publishing that bound next to the metric costs one line, requires no reader and no new acquisition, and makes explicit what the metric is silent about. It is the cheapest useful step towards evaluation that could detect the failure this review is concerned with.

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Ethics statement

This review synthesises secondary data from published, de-identified study reports. It involved no human participants, no animal experimentation and no collection of private health data, so institutional review board approval and patient informed consent were not required.

Data and code availability

The review protocol, executed search strategy, screening criteria, every title/abstract and full-text decision with its recorded reason, the completed PRISMA 2020 checklist, the per-study extraction and QUADAS-2 worksheets for all 263 included studies, the analysis code, and the duplicate-screening sample and per-reviewer appraisal worksheet with their agreement computations (`kappa_worksheet_214_HUMAN_*.xlsx`, `kappa_human_results.csv` and `quadas2_three_reviewer_worksheet_18.csv`) are openly available on the Open Science Framework at `https://doi.org/10.17605/OSF.IO/EUXA8` and mirrored at `https://github.com/dmai287/brain-mri-recon-safety-slr`, archived at Zenodo under DOI `10.5281/zenodo.21845858` (release `v1.0.2`, commit `219242a`; concept DOI `10.5281/zenodo.21836966`). Every count reported here is regenerable from `included_characteristics.csv` in that release, and the deposit encodes the same reader/algorithmic partition (57/206) that this article reports, with the four reclassified studies flagged individually. The two figures generated from that file and the ΔPSNR identity figure are produced by `5_Code/make_figures.py` and `5_Code/make_dpsnr_figure.py`. The deposit is retrospective and is documented as such. Full-text PDFs of the included studies are not redistributed, as publisher licence terms do not permit it; the DOI list in the released bibliography (`included_studies_263.bib`) identifies the corpus in full.

CRediT authorship contribution statement

Dat Tat Mai: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Project administration, Writing – original draft, Writing – review & editing. **Thai Viet Pham:** Investigation, Data curation, Validation, Resources, Writing – review & editing. **Thu Nguyen Thi Dang:** Investigation, Data curation, Validation, Writing – review & editing. **James Jin Kang:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used generative AI technologies solely to assist with language editing and proofreading of the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All study screening, data extraction, and quality appraisals reported in this review were conducted manually by the authors.

Appendix A. The included studies and how to trace a claim to them

All 263 included primary studies appear in the reference list, so the evidence base is part of the article of record rather than of the supplement alone. Because the synthesis is narrative and most claims are grouped rather than attached to a single study, the reference list alone does not show which studies produced a given number. That mapping is carried by `included_characteristics.csv`, which holds one row per included study with its extracted method family, evaluation paradigm, failure modes, sequences, field strength, reference standard and code availability, each with the verbatim quotation supporting it, so any count, share or crosstab in Section 4 can be reproduced and the contributing studies listed by filtering that file. Supplementary Tables S1–S3 give the same 263 studies as a numbered index by publication year, and `included_characteristics_supplement.csv` adds the per-study QUADAS-2 and reproducibility ratings.

The reference list therefore contains two kinds of entry, and a reader needs a rule to tell them apart. It prints 299: the 263 included studies, carried into the list under PRISMA 2020 item 17, and 36 narrative-only works cited in argument. **An entry appearing in the numbered index of Supplementary Tables S1–S3 is a member of the reviewed corpus; an entry that does not is a narrative citation.** Equivalently, and checkable without the supplement, every narrative citation is cited at a specific point in the running text of Sections 1–6, whereas corpus entries not discussed by name appear without an in-text citation. The two lists were generated by independent mechanisms and agree exactly as sets. We would be glad to move the corpus to a separately labelled supplementary reference list if the editor prefers that production route.

Appendix B. Supplementary material

Supplementary material associated with this article (Supplementary Tables S1–S12 and Supplementary Figure S1) is provided as a separate document. Tables S1–S3 list all 263 included primary studies with full bibliographic details;

Tables S4–S9 give the comparison against related secondary studies, the search vocabulary and its derivation, the extraction form, the thematic evidence architecture, the four metric-failure mechanisms with their evidence sources, and the provenance of every released data file. Tables S10–S12 carry the supporting detail for Section 4.5: the annotation-layer derivation for fastMRI+ (S10), the disjoint-strata correction for the sequence contrast (S11), and the venue cross-tabulation with evaluation-paradigm multiplicity by era (S12).

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